Contents

Chapter 1

Graphing Polynomials

Polynomials are used to solve real-world problems! To do this, we need to know what these polynomials look like on a graph. You have many tools you can use to sketch out graphs. These graphs are simple visuals that help to evaluate several different points on the plot.

For instance, you can determine:

- where the graph crosses the y-axis, which you can find by evaluating the polynomial at $x = 0$.

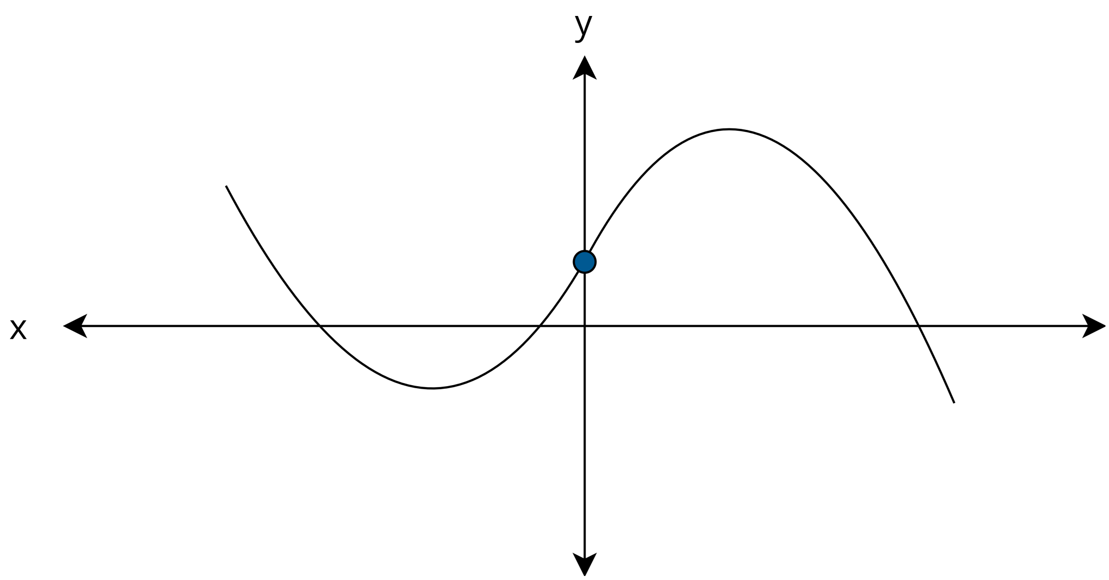


Figure 1.1: A graph showing the y-intercept of a polynomial

- where the graph crosses the x-axis, which you can find by solving for the roots of the polynomial.

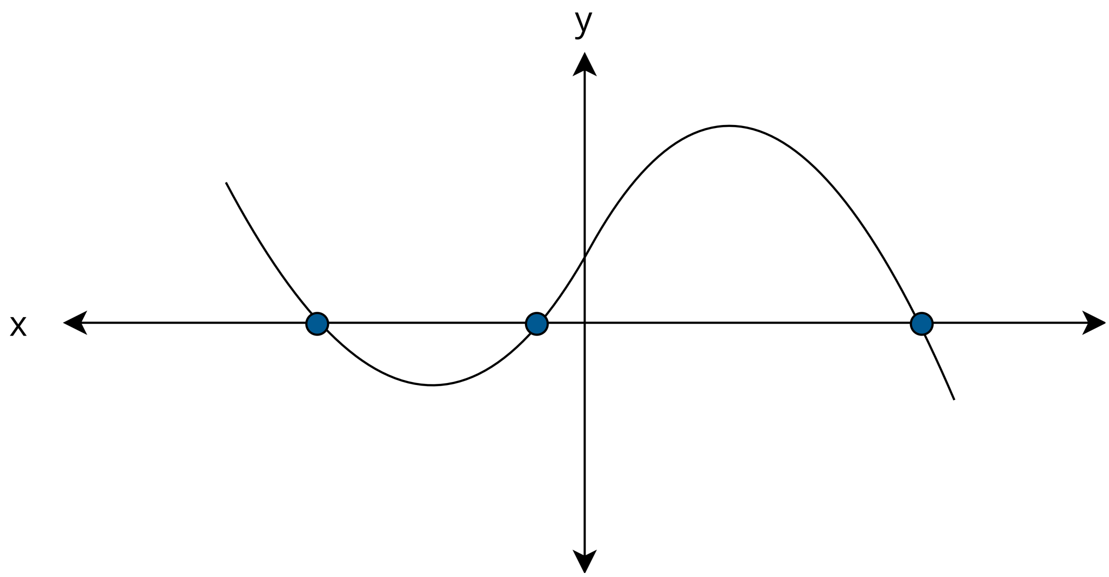


Figure 1.2: A graph showing the roots of a polynomial

- the level spots on the graph (often the top of a hump or the bottom of a dip). You can take the derivative of the polynomial (which is also a polynomial), and find the roots of that.

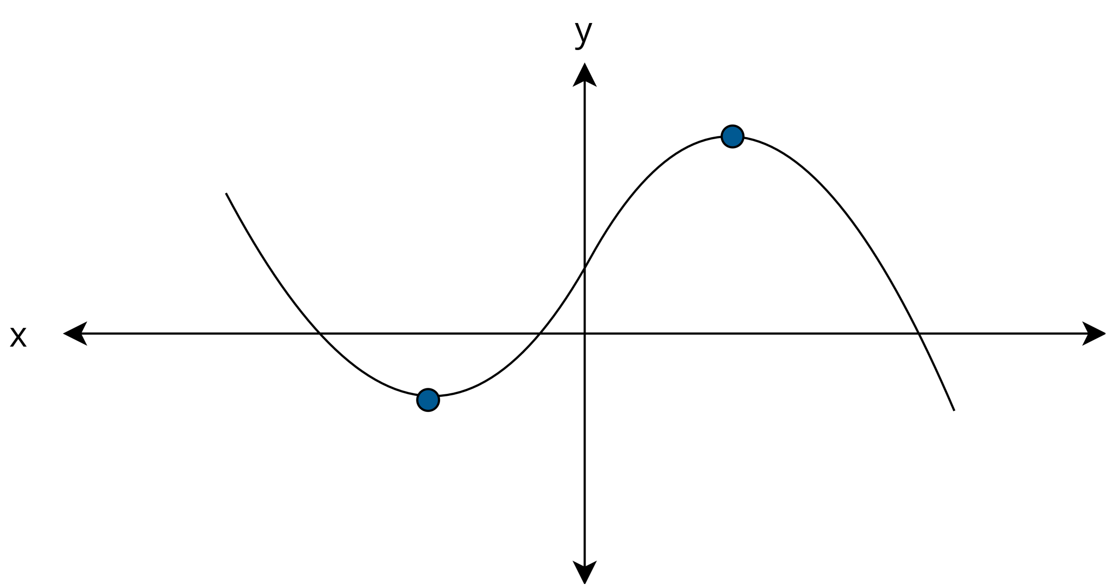


Figure 1.3: A graph showing "level spots" of a polynomial.

For example, if you wanted to graph the polynomial $f(x) = -x^3 - x^2 + 6x$, you might plug in a few values that are easy to compute:

- $f(-2) = -8$
- $f(-1) = -6$
- $f(0) = 0$
- $f(1) = 4$
- $f(2) = 0$

So, right away, we know two roots: $x = 0$ and $x = 2$. Are there others? We won't know until we factor the polynomial:

$$
\begin{aligned}
&-x^3 - x^2 + 6x \\
&= (-1x)(x^2 + x - 6) \\
&= (-1x)(x + 3)(x - 2)
\end{aligned}
$$

So, yes, there is a third root: $x = -3$

What about the level spots? $f'(x) = -3x^2 - 2x + 6$. Where is that zero?

$$
\begin{aligned}
&-3x^2 - 2x + 6 = 0 \\
&x^2 + \frac{2}{3}x - 2 = 0
\end{aligned}
$$

We have a formula for quadratics like this:

$$
\begin{aligned}
x &= -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2} \\
&= -\frac{\frac{2}{3}}{2} \pm \frac{\sqrt{\left(\frac{2}{3}\right)^2 - 4(-2)}}{2} \\
&= -\frac{1}{3} \pm \frac{\sqrt{\frac{4}{9} + 8}}{2} \\
&= -\frac{1}{3} \pm \frac{\sqrt{\frac{85}{9}}}{2} \\
&= -\frac{1}{3} \pm \frac{\sqrt{85}}{6} \\
&\approx 1.20 \text{ and } -1.87
\end{aligned}
$$

Now, you might plug those numbers in:

- $f(1.2) \approx 4.0$
- $f(-1.87) \approx -8.2$

1.1 Leading term in graphing

There is one more trick you need before you can draw a good graph of a polynomial. As you go father and farther to the left and right, where does the function go? In other words, does the graph go up on both ends (like a smile)? Or does it go down on both ends (like a frown)? Or does the negative end go down (frowny) while the positive end go up (smiley)? Or does the negative go up (smiley) and the positive end go down (frowny)?

Assuming the polynomial is not constant, there are only those four possibilities. It is determined entirely by the leading term of the polynomial. If the degree of the leading term is even, both ends go in the same direction (both are smiley or both are frowny). If the coefficient of the leading term is positive, the positive end is smiley.

The graph we are working on has a leading term of $-1x^3$. The degree is odd, thus the ends go in different directions. The coefficient is negative, so the positive end points down. Now, you can draw the graph, which should look something like this:

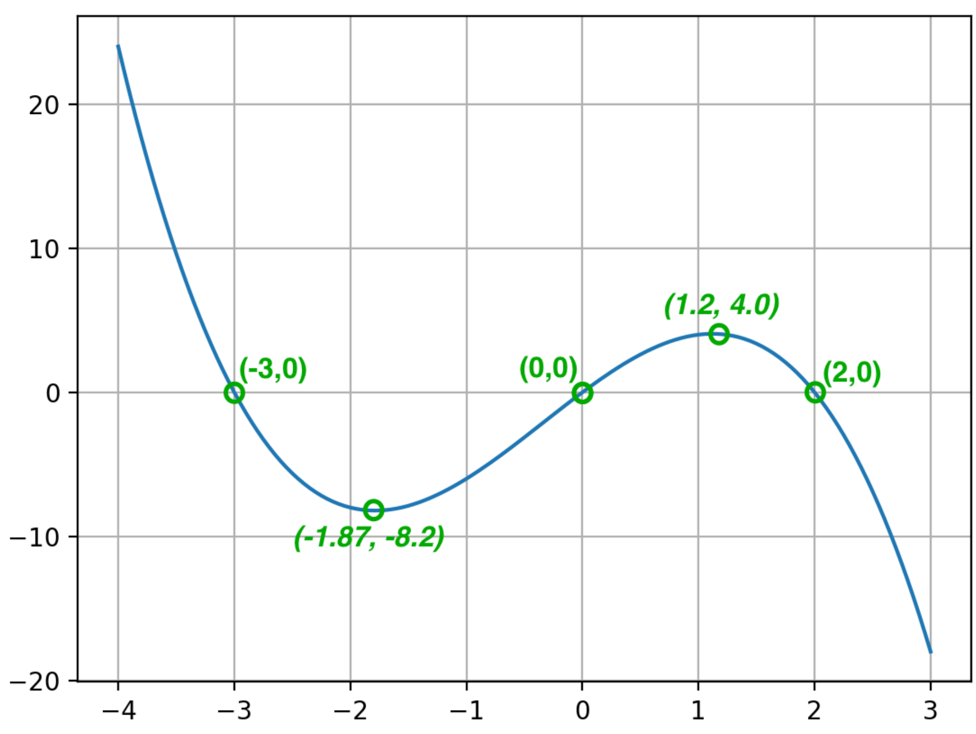


Figure 1.4: A graph with roots and significant points.

Chapter 2

Interpolating with Polynomials

Let's say someone on a distant planet records video of a hammer being thrown up into the air. They send you three random frames of the hammer in flight. Each frame has a timestamp, and you can clearly see how high the hammer is in each one. Can you create a 2nd degree polynomial that explains the entire flight of the hammer?

In other words, you have three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$. Can you find a, b, c such that the graph of $at^2 + bt + c = t$ passes through all three points?

The answer is yes. In fact, given any n points[1], there is exactly one $n-1$ degree polynomial that passes through all the points.

There are a lot of variables floating around. Let's make it concrete: The photos are taken at $t = 2$ seconds, $t = 3$ seconds, and $t = 4$ seconds. In those photos, the height of the hammer is $5m$, $7m$, and $6m$. So, we want our polynomial to pass through these points: (2, 5), (3, 7), (4,6).

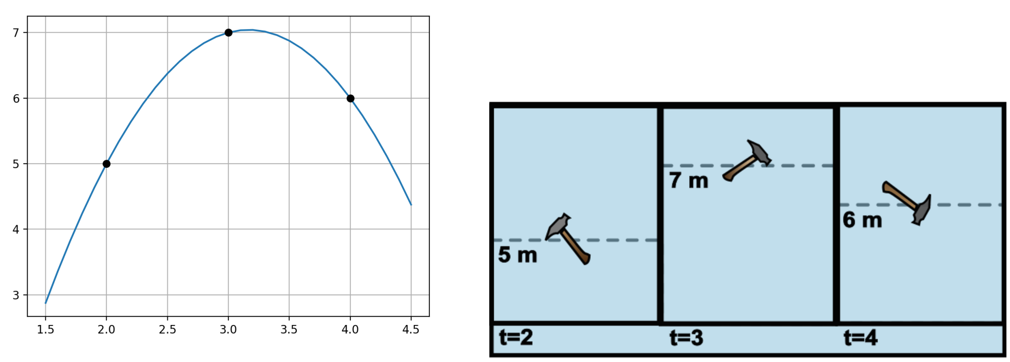


Figure 2.1: Left: the graph of the hammer going through the given points. Right: the three hammer frames.

How can you find that polynomial? Let's do it in small steps. Can you create a 2nd degree polynomial that is not zero at $t = 2$, but is zero at $t = 3$ and $t = 4$? Yes, you can: $(x-3)(x-4)$ has exactly two roots at $t = 3$ and $t = 4$. The value of this polynomial at $t = 2$ is $(2 - 3)(2 - 4) = 2$. We really want it to be $5m$, so we can divide the whole polynomial by 2 and multiply it by 5.

[1] $n = 2$ will result in a line in slope intercept point, and $n = 1$ is a single point

Now we have the polynomial:

$$f_0(x) = \frac{5}{(2-3)(2-4)}(x-3)(x-4) = \frac{5}{2}x^2 - \frac{35}{2}x + 30$$

This is a second degree polynomial that is 5 at $t = 2$ and 0 at $t = 3$ and $t = 4$.

Now, we create a polynomial that is 7 at $t = 3$ and 0 at $t = 2$ and $t = 4$:

$$f_1(x) = \frac{7}{(3-2)(3-4)}(x-2)(x-4) = -7x^2 + 42x - 56$$

Finally, we create a polynomial that is 6 at $t = 4$ and zero at $t = 2$ and $t = 3$:

$$f_2(x) = \frac{6}{(4-2)(4-3)}(x-2)(x-3) = 3x^2 - 15x + 18$$

Adding these three polynomials together gives you a new polynomial that touches all three points:

$$f(x) = \frac{5}{2}x^2 - \frac{35}{2}x + 30 - 7x^2 + 42x - 56 + 3x^2 - 15x + 18 = -\frac{3}{2}x^2 + \frac{19}{2}x - 8$$

You can test this in Python with your `Polynomial` class. Create a file called `test_interpolation.py`. Add this code:

```python
from Polynomial import Polynomial
import matplotlib.pyplot as plt

in_x = [2,3,4]
in_y = [5,7,6]

pn = Polynomial([-8, 19/2, -3/2])
print(pn)

# These lists will hold our x and y values
x_list = []
y_list = []

# Starting x
current_x = 1.5

while current_x <= 4.5:
    # Evaluate pn at current_x
    current_y = pn(current_x)

    # Add x and y to respective lists
    x_list.append(current_x)
```

```
    y_list.append(current_y)

    # Move x forward
    current_x += 0.05

# Plot the curve
plt.plot(x_list, y_list)

# Plot black circles on the given points
plt.plot(in_x, in_y, "ko")
plt.grid(True)
plt.show()
```

You should get a nice plot that shows the graph of the polynomial passing through those three points.

In general, then, if you provide any three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$, there is a second degree polynomial that pass through all three:

$$\frac{h_0}{(t_0 - t_1)(t_0 - t_2)}(x - t_1)(x - t_2) + \frac{h_1}{(t_1 - t_0)(t_1 - t_2)}(x - t_0)(x - t_2) + \frac{h_2}{(t_2 - t_0)(t_2 - t_1)}(x - t_0)(x - t_1)$$

What if you are given 9 points $((t_0, h_0), (t_1, h_1), \ldots, (t_8, h_8))$ and want to find a 8th degree polynomial that passes through all of them? Just what you would expect:

$$\frac{h_0}{(t_0 - t_1)(t_0 - t_2)\ldots(t_0 - t_8)}(x - t_1)(x - t_2)\ldots(x - t_8) + \ldots + \frac{h_8}{(t_8 - t_0)\ldots(t_8 - t_7)}(x - t_0)\ldots(x - t_7)$$

FIXME: Do I need to define summation and prod here?

The general solution is, given n points, the $n - 1$ degree polynomial that goes through them is

$$y = \sum_{i=0}^{n} \left(\prod_{\substack{0 \le j \le n \\ j \ne i}} \frac{x - t_j}{t_i - t_j} \right) h_i$$

That would be tedious for a person to compute, but computers are perfect for this stuff. Let's create a method that creates instances of Polynomial using interpolation.[2]

[2]Note that other textbooks or resources may refer to this as *Lagrange Polynomial Interpolation*.

2.1 Interpolating polynomials in python

Your method will take two lists of numbers: one contains x-values and the other contains y-values. So comment out the line that creates the polynomial in `test_interpolation.py` and create it from two lists:

```
in_x = [2,3,4]
in_y = [5,7,6]
# pn = Polynomial([-8, 19/2, -3/2])
pn = Polynomial.from_points(in_x, in_y)
print(pn)
```

Add the following method to your Polynomial class in `Polynomial.py`

```
@classmethod
    def from_points(cls, x_values, y_values):
        coef_count = len(x_values)

        # Sums start with a zero polynomial
        sum_pn = Polynomial([0.0] * coef_count)
        for i in range(coef_count):

            # Products start with the constant 1 polynomial
            product_pn = Polynomial([1.0])
            for j in range(coef_count):

                # Must skip j=i
                if j != i:
                    # (1x - x_values[j]) has a root at x_values[j]
                    factor_pn = Polynomial([-1 * x_values[j], 1])
                    product_pn = product_pn * factor_pn

            # Scale so product_pn(x_values[i]) = y_values[i]
            scale_factor  = y_values[i] / product_pn(x_values[i])
            scaled_pn = scale_factor * product_pn

            # Add it to the sum
            sum_pn = sum_pn + scaled_pn

        return sum_pn
```

It should work exactly the same as before. You should get the same polynomial printed out as before as well as the same plot of the curve passing through the three points.

How about five points? Change `in_x` and `in_y` at the start of `test_interpolation.py`:

```
in_x = [1.7, 2, 2.7, 3.5, 4, 4.4]
```

```
in_y = [8, 12, 1, 4, -1, 6]
```

You should get a polynomial that passes through all five points:

$$11.21x^5 - 171.05x^4 + 1019.44x^3 - 2957.53x^2 + 4161.78x - 2258.75$$

It should look like this:

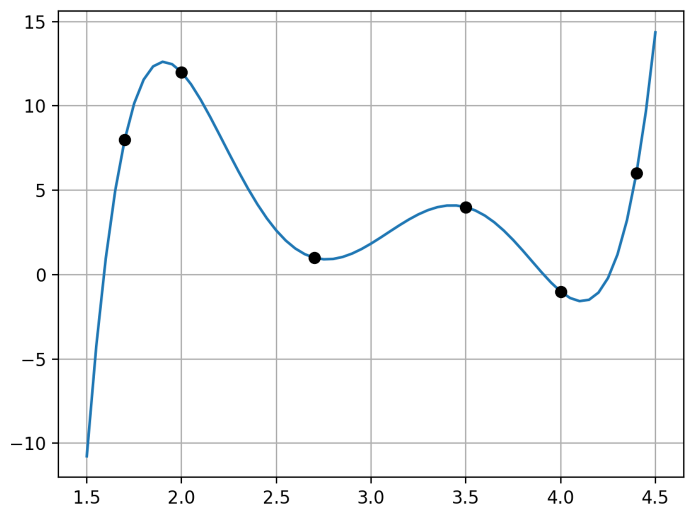


Figure 2.2: 5 point interpolation of points.

Chapter 3

Rational Functions

We have discussed addition, subtraction, and multiplication of polynomials. What about division?

A quotient of polynomials is called a rational expression. When the polynomials are factored and the stars align, we can simplify the rational expression to a single polynomial, just like we might reduce a fraction to lowest terms.

Example

$$\begin{aligned} \frac{(x+1)(x+5)}{x+5} &= (x+1) \cdot \frac{x+5}{x+5} \\ &= x+1 \qquad (x \neq 5) \end{aligned} \tag{3.1}$$

[1]

What if the polynomials are not factored? Factor them first.

Example

$$\frac{x^2+6x+5}{x+5} = \frac{(x+1)(x+5)}{x+5}$$

and simplify as in the previous example.

Now, let us consider a rational expression which can be simplified to a single polynomial — but in the denominator.

Example

$$\begin{aligned} \frac{x+5}{x^2+6x+5} &= \frac{x+5}{(x+1)(x+5)} \\ &= \frac{1}{x+1} \cdot \frac{x+5}{x+5} \\ &= \frac{1}{x+1} \end{aligned} \tag{3.2}$$

Consider this expression as a function: $f(x) = \frac{1}{x+1}$. As you might have guessed, this is

[1] It is important to note that we need to restrict $x \neq 5$ since substituting $x = 5$ would produce division by zero, making the expression undefined. Canceling the $(x+5)$ factor is only valid when $x \neq 5$, so the simplified form $x+1$ and the original expression are equal everywhere *except* at $x = 5$.

called a rational function. We did not bother looking at the result of the previous example as a function, because we already know that function type: it is a line with slope 1 and y-intercept 1. However, this rational function is another animal entirely. Let us examine our first rational function with a familiar concept: the y-intercept.

y-intercept: $f(0) = \frac{1}{0+1} = \frac{1}{1} = 1$. The graph contains the point $(0, 1)$.

Does f have an x-intercept? That would be an x-value where $f(x) = 0$. But a fraction equals 0 only when its numerator equals 0; since the numerator of this expression is always 1, f has no x-intercept.

Knowing the y-intercept, and that there is no x-intercept, is a comforting start. But things get weird when we consider a concept that has previously seemed quite simple: domain. Recall that the domain of a function is the set of all values which can be used as inputs. In this case, the domain includes all real numbers, with one exception. The number -1 is not a valid input because $f(-1) = \frac{1}{-1+1} = \frac{1}{0}$, which is undefined. So, we say that the domain is all real numbers except -1. This means the graph contains a point corresponding to every x-value except -1.

There is no point at $x = -1$, but there is a point at every other x-value, such as, for example, -1.1, or -0.99999. So, what is happening near $x = -1$?

x	-1.1	-1.01	-1.001	-0.999	-0.99	-0.9
f(x)	-10	-100	-1000	1000	100	10

The function is going haywire. As we choose x-values closer and closer to -1, the resulting function values are larger and larger in magnitude. Also, they are negative on one side, but positive on the other. So, how does a graph go from y-values of -10, to -100, to -1000, all in a space of less than 0.1 on the x-axis? Then, from there, suddenly to big positive numbers on the other side of $x = -1$? All without ever crossing the x-axis (since there is no x-intercept)? Let's look at the graph.

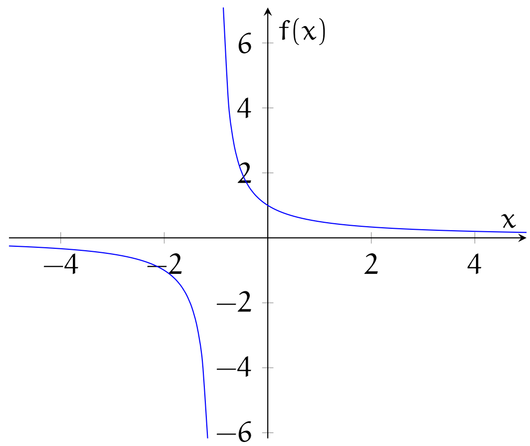


Figure 3.1: Graph of $f(x) = \frac{1}{x+1}$

We can see the y-intercept we found above. We can also see that the graph has no x-intercept, as expected. The phenomenon occurring at $x = -1$ is called a vertical asymptote. One other interesting feature of this graph is how it hugs the x-axis toward the left and right edges of the window. This makes the line $y = 0$ (the x-axis) a horizontal asymptote for this function. We can see why this is happening numerically by considering what happens for x-values far from 0. In this function, the result is a fraction with a numerator of 1 and a denominator that is large in size: a fraction that is close to 0.

x	-1000	-100	-10	10	100	1000
$f(x)$	-0.001	-0.01	-0.1	0.1	0.01	0.001

Let's examine another rational function. Begin by factoring to see if the function can be simplified.

$$g(x) = \frac{x^2 - 3x + 2}{x^2 - 4x + 3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$$

Consider the domain of g before continuing. Which values of x are valid inputs? Since substituting $x = 1$ or $x = 3$ would result in division by 0, these are not valid inputs. The domain of g is all real numbers except 1 and 3.

Now, for any x-value except 1, $\frac{x-1}{x-1} = 1$. This means that, for all x-values but 1, we can cancel those factors, leaving $g(x) = \frac{x-2}{x-3}$. (We will talk more about what is happening at $x = 1$ in a moment.)

This function has both x- and y-intercepts: y-intercept: $g(0) = \frac{0-2}{0-3} = \frac{2}{3}$. The graph contains the point $(0, \frac{2}{3})$. x-intercept: $g(x) = 0$ where the numerator equals 0 and the denominator does not equal 0. Since $x - 2 = 0$ when $x = 2$, the x-intercept is 2 and the graph contains the point $(2, 0)$.

The graph of g has a vertical asymptote at any x-value where substitution would result in dividing a nonzero number by zero. Thus, g has a vertical asymptote at $x = 3$.

Does g have a horizontal asymptote? Let us see what happens when we substitute x-values far from 0.

x	-1000	-100	-10	10	100	1000
$g(x)$	0.999	0.990	0.923	1.143	1.010	1.001

As we move further away from the y-axis, the y-values become closer to 1. The horizontal asymptote describes the end behavior of the function, or what the graph looks like far from the y-axis. In this case, if we ignore the portion close to the y-axis, the graph begins to look like the line $y = 1$, making this the horizontal asymptote of g.

So, what is happening at $x = 1$? The value is not in the domain of the function, but there is no vertical asymptote there. That is because substituting any other value for x, even values very close to 1, into $\frac{(x-1)(x-2)}{(x-1)(x-3)}$ gives the exact same number as substituting into $\frac{x-2}{x-3}$. So, there is a hole in the graph at $x = 1$, but nothing strange is happening on either side of 1. (Depending on the graphing software, the hole may not be visible.)

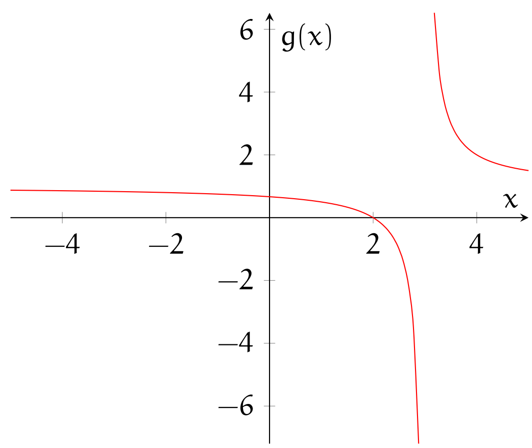


Figure 3.2: Graph of $g(x) = \frac{x^2-3x+2}{x^2-4x+3}$

Exercise 1 Rational Functions Practice 1

Determine the x- and y-intercepts and horizontal and vertical asymptotes of the rational function:

1. $\frac{2x+5}{x+4}$

Working Space

Answer on Page 113

Exercise 2

[This question was originally presented in the no-calculator section of the 2012 AP Calculus BC exam.] The line $y = 5$ is a horizontal asymptote of which of the following functions? Explain.

A. $y = \frac{\sin 5x}{x}$
B. $y = 5x$
C. $y = \frac{1}{x-5}$
D. $y = \frac{5x}{1-x}$
E. $y = \frac{20x^2-x}{1+4x^2}$

Working Space

Answer on Page 113

In those examples, common factors cancel, leaving one polynomial. Of course, there is no guarantee that any two polynomials will have common factors, or even be factorable at all. Now, we consider an example that cannot be simplified. We will focus on just the asymptotes here.

$$h(x) = \frac{x^2}{x-1}$$

We see that the x-value 1 gives division of a non-zero number by zero, giving a vertical asymptote at $x = 1$. How about a horizontal asymptote? We examine values of h for values of x far from 0.

x	-1000	-100	-10	10	100	1000
$h(x)$	-999	-99	-9	11	101	1001

Rather than seeing function values leveling off as in the previous examples, we see function values that grow in size along with x. The function h has no horizontal asymptote. Let's examine the graph:

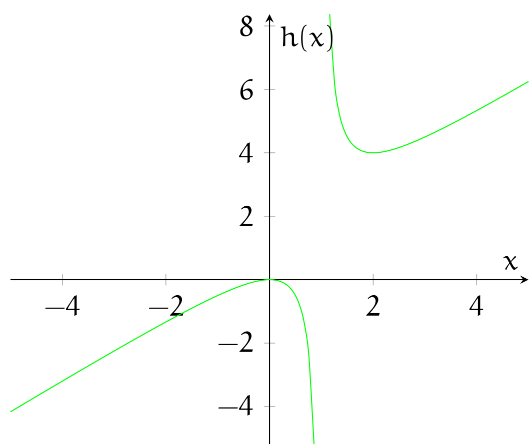


Figure 3.3: Graph of $h(x) = \frac{x^2}{x-1}$

This function exhibits a different type of end behavior: that of a line with slope 1. To see that, cover up the portion of the graph near the y-axis and focus on the left and right. The rather dull and time-consuming technique of polynomial long division can be used to rewrite the function as a quotient and a remainder. We encourage you to watch the Khan Academy video on the topic, but for now, let us instead use our knowledge of factoring techniques and a clever little trick.

$$
\begin{aligned}
h(x) &= \frac{x^2}{x-1} \\
&= \frac{x^2-1+1}{x-1} \\
&= \frac{x^2-1}{x-1} + \frac{1}{x-1} \\
&= \frac{(x-1)(x+1)}{x-1} + \frac{1}{x-1} \\
&= x+1+\frac{1}{x-1}
\end{aligned}
\tag{3.3}
$$

We obtain a quotient of $x+1$ and a remainder of 1. It is the quotient that determines the end behavior of the graph. Why? Substituting x-values far from zero makes the remainder term very small, since it becomes a fraction with a large denominator but a numerator of only 1. So for x-values far from zero, the y-value is x plus 1 plus a very small number (so small that we can justifiably ignore it). This means that far from the y-axis, the function acts like the quotient: the line $y = x+1$. We call this line an oblique asymptote. See below how the graph of $h(x)$ hugs that line.

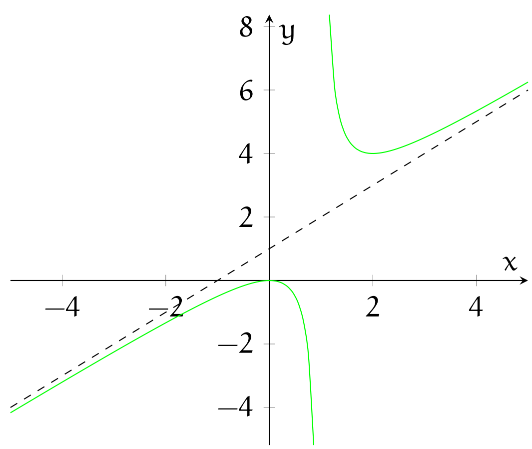


Figure 3.4: Graph of $h(x) = \frac{x^2}{x-1}$ and its oblique asymptote $y = x+1$

Exercise 3 Rational Functions Practice 2

Factor and simplify the rational function, then determine any holes and vertical and oblique asymptotes of the rational function.

1. $\frac{x^3+2x^2}{x^2+x}$

Working Space

Answer on Page 113

We have seen lines act as end behaviors. Are there other possibilities? Sure! Here is an example with parabolic end behavior.

$$k(x) = \frac{x^3}{x-2}$$

We use our add-subtract trick to reveal the quotient, which describes the end behavior.

$$\begin{aligned} h(x) &= \frac{x^3}{x-2} \\ &= \frac{x^3-8+8}{x-2} \\ &= \frac{x^3-8}{x-2} + \frac{8}{x-2} \\ &= \frac{(x-2)(x^2+2x+4)}{x-2} + \frac{8}{x-2} \\ &= x^2+2x+4+\frac{8}{x-2} \end{aligned} \tag{3.4}$$

The quotient, x^2+2x+4, should describe the end behavior. We confirm by graphing both

k and the quotient - the parabolic asymptote.

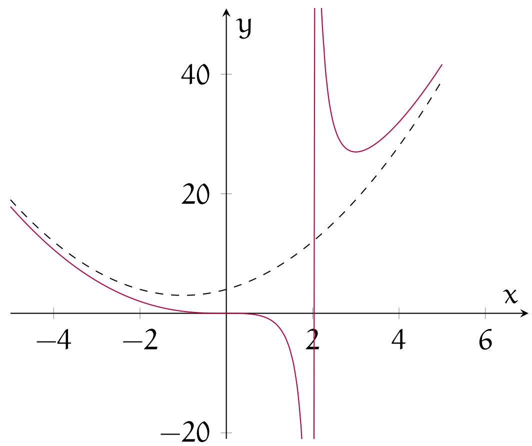


Figure 3.5: Graph of $k(x) = \frac{x^3}{x-2}$ and its parabolic asymptote $y = x^2 + 2x + 4$

Chapter 4

Partial Fractions

How can you add fractions with different denominators, like $\frac{1}{x} + \frac{2}{x+3}$? You would need to make the denominators the same; after that, you could just add the numerators. You achieve this by multiplying the numerator and denominator of each fraction by the denominator of the other fraction:

$$\frac{1}{x} + \frac{2}{x+3} = \frac{1}{x}\left(\frac{x+3}{x+3}\right) + \frac{2}{x+3}\left(\frac{x}{x}\right)$$

Recall that when the numerator and denominator of a fraction are the same, the fraction is equal to one. So, we are not changing the *value* of each fraction, since we are just multiplying by one. Continuing, we can perform the multiplication and see that:

$$\frac{1}{x} + \frac{2}{x+3} = \frac{x+3}{x(x+3)} + \frac{2x}{x(x+3)}$$

$$= \frac{(x+3)+2x}{x(x+3)} = \frac{3x+3}{x^2+3x} = \frac{3(x+1)}{x^2+3x}$$

The inverse of this process is called **partial fraction decomposition** (or partial fraction expansion). This method has applications in many fields, but we will find it most useful as a tool to evaluate integrals in a later chapter.

Let $g(x)$ be a rational function such that

$$g(x) = \frac{P(x)}{Q(x)}$$

Where $P(x)$ and $Q(x)$ are polynomials. If $g(x)$ is proper (that is, the degree of P is less than the degree of $Q(x)$ then we can express $g(x)$ as the sum of simpler rational fractions. If $g(x)$ is improper (that is, the degree of P is greater than or equal to the degree of Q), then we must first perform long division to obtain a remainder, $R(x)$, where the degree of R is less than the degree of Q:

$$g(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

4.0.1 Improper fractions

What is $\int \frac{x^3+x}{x-1}\,dx$. Using long division, we see that:

$$\frac{x^3+x}{x-1} = x^2 + x + 2 + \frac{2}{x-1}$$

(see figure 4.1 for an explanation). Then we can also say that:

$$\frac{x^3+x}{x-1} = x^2 + x + 2 + \frac{2}{x-1}$$

$$\begin{array}{r l}
 & \;x^2 + x + 2 \\ \cline{2-2}
x - 1 & \big)\; x^3 + 0x^2 + x \\
 & \underline{-(x^3 - x^2)} \\
 & \qquad x^2 + x \\
 & \qquad \underline{-(x^2 - x)} \\
 & \qquad\qquad 2x \\
 & \qquad\qquad \underline{-(2x - 2)} \\
 & \qquad\qquad\quad 2
\end{array}$$

Figure 4.1: Evaluating $(x^3 + x) \div (x - 1)$ with the long division method

When you start with an improper fraction, use long division to reduce it to a term plus a proper fraction, then use the methods outlined below to further manipulate the proper fraction.

Exercise 4

Use long division to reduce the following improper rational functions to a term plus a proper rational fraction.

1. $\frac{x^4+x^3+2x^2+2x-3}{x^2-3x+2}$
2. $\frac{2x^3+5}{x^3-3x^2+2x-4}$
3. $\frac{3x^4-2x^3-x^2+1}{x^3-3x}$

Working Space

Answer on Page 113

4.0.2 Proper fractions

When the order of the numerator is less than or equal to the denominator, there are three further possibilities.

No repeated linear factors

In the first case, the denominator, $Q(x)$ is composed of distinct linear factors. In this case, we can say that $Q(x) = (a_1x + b_1)(a_2x + b_2)\cdots(a_nx + b_n)$, where no factor is repeated (including constant multiples). Then, there exists $A, B, C, \cdots$, such that:

$$\frac{P(x)}{Q(x)} = \frac{A}{a_1x + b_1} + \frac{B}{a_2x + b_2} + \cdots$$

Let's see an example of this by decomposing $\frac{4x^2-7x-12}{x(x+2)(x-3)}$. We start by defining A, B, and C, such that:

$$\frac{4x^2 - 7x - 12}{x(x+2)(x-3)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-3}$$

Multiplying both sides by $x(x + 2)(x - 3)$ we get:

$$4x^2 - 7x - 12 = A(x + 2)(x - 3) + B(x)(x - 3) + C(x)(x + 2)$$

We have 3 unknowns and only one equation! Don't worry — remember this equation is true for all x, so we can choose a convenient value of x to isolate each unknown in turn. Starting, let $x = 0$. Then:

$$4(0)^2 - 7(0) - 12 = A(0+2)(0-3) + B(0)(x-3) + C(0)(x+2)$$

$$-12 = A(2)(-3) + 0 + 0$$

Notice that the B and C disappear, and we can solve for A:

$$A = \frac{-12}{-6} = 2$$

We can solve for B by setting $x = -2$ and for C by setting $x = 3$ (notice, we've used all three zeroes of the denominator polynomial):

$$4(-2)^2 - 7(-2) - 12 = A(-2+2)(-2-3) + B(-2)(-2-3) + C(-2)(-2+2)$$

$$4(4) + 14 - 12 = 0 + B(-2)(-5) + 0$$

$$16 + 2 = 10B$$

$$B = \frac{9}{5}$$

and

$$4(3)^2 - 7(3) - 12 = A(3+2)(3-3) + B(3)(3-3) + C(3)(3+2)$$

$$4(9) - 21 - 12 = 0 + 0 + C(3)(5)$$

$$36 - 33 = 15C$$

$$C = \frac{1}{5}$$

We can then decompose our original fraction:

$$\frac{4x^2 - 7x - 12}{x(x+2)(x-3)} = \frac{2}{x} + \frac{9}{5(x+2)} + \frac{1}{5(x-3)}$$

You can check your answer by cross-multiplying and adding. You should get the same rational function back.

Repeated linear factors

The second case is if $Q(x)$ has repeated factors (such as $x^2 + 8x + 16 = (x+4)^2$). Suppose the first linear factor, $(a_1x + b_1)$ is repeated r times (that is, $Q(x)$ contains the factor $(a_1x + b_1)^r$). Then, instead of $\frac{A}{a_1x+b_1}$, we should write:

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

Let's look at a concrete example to see how this works:

Example: Decompose $\frac{x^2+x+1}{(x+1)^2(x+2)}$

Solution: We start by defining:

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$$

Multiplying both sides by $(x+1)^2(x+2)$:

$$x^2+x+1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$$

Since there are only 2 roots to $(x+1)^2(x+2)$, we will use another method called "equating the coefficients" to find A, B, and C. We start by expanding the right side of the equation:

$$x^2+x+1 = A(x^2+3x+2) + B(x+2) + C(x^2+2x+1)$$

Distributing and combining, we find that:

$$x^2+x+1 = Ax^2+3Ax+2A+Bx+2B+Cx^2+2Cx+C$$

$$x^2+x+1 = (A+C)x^2+(3A+B+2C)x+(2A+2B+C)$$

For this equation to be true, we know that:

$$A+C=1$$

$$3A+B+2C=1$$

$$2A+2B+C=1$$

(That is, the coefficient for x^2 on the left, 1, must be equal to the coefficient for x^2 on the right, $(A+C)$, and so on.) We now have a system of 3 equations and 3 unknowns. When you solve for each, you should find that:

$$A=-2$$

$$B=1$$

$$C=3$$

Therefore,

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$$

Irreducible quadratic factors

Sometimes, we cannot express a polynomial as the product of two linear statements (that is, terms in the form $ax+b$). Take x^2+1, which cannot be expressed as the product of real, linear terms. What do you do if something like x^2+1 is in the denominator? In this case, when we write an expression for $\frac{P(x)}{Q(x)}$, we include a term in the form:

$$\frac{Ax + B}{ax^2 + bx + c}$$

For example, we can write:

$$\frac{x}{(x-2)(x^2+1)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{x^2+4}$$

Example: Decompose $\frac{2x^2-x+4}{x^3+4x}$

Solution: We begin by factoring the denominator:

$$x^3 + 4x = x(x^2 + 4)$$

Which cannot be factored further. Therefore, we define:

$$\frac{2x^2 - x + 4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

$$2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x$$

$$2x^2 - x + 4 = Ax^2 + 4A + Bx^2 + Cx$$

Which implies that:

$$2 = A + B$$

$$C = -1$$

$$4A = 4$$

Therefore, $A = 1$, $B = 1$, and $C = -1$ and we can say that:

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{1}{x} + \frac{x - 1}{x^2 + 4}$$

Repeated irreducible quadratic factors

Lastly, the denominator might contain repeated irreducible quadratic factors. Similar to repeated linear factors, when setting up your partial fractions, instead of only writing

$$\frac{A}{ax^2 + bx + c}$$

For a quadratic factor that is repeated r times, your equation should include:

$$\frac{A_1}{ax^2 + bx + c} + \frac{A_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_r}{(ax^2 + bx + c)^r}$$

Exercise 5

Decompose the following proper fractions

1. $\frac{x-4}{x^2+5x-6}$
2. $\frac{x^2+x+1}{(x^2+1)^2}$
3. $\frac{x^2+x+1}{(x+1)^2(x+2)}$

Working Space

Answer on Page 113

CHAPTER 5

Limits

5.1 Introduction to Limits

To begin talking about limits, let's imagine a a scenario:

I flip a coin 10 times, and count the number of times it lands on heads. Let's say we get the following sequence

$$H, T, T, H, T, H, T, T, H, T$$

the fraction of heads to total flips is

$$\frac{\text{number of heads}}{\text{number of flips}} = \frac{4}{10} = 0.4$$

Now, if we were to flip the coin 100 times instead, and, we got 46 heads. Our sample proportion gets closer to 0.5, which is our *expected value* or *mean*:

$$\frac{\text{number of heads}}{\text{number of flips}} = \frac{46}{100} = 0.46$$

As we the number flips increases, the fraction of heads to total flips approaches the true probability of heads. Flipping the coin $10,000$ times would produce a result closer to the expected value, such as $5,230$ making a fraction of 0.523. While all of the flips are individual events, the long running average stabilizes closer to the true probability.

This long-run stabilization is one version of the *Law of Large Numbers*, an important idea in statistics.

We will revisit this in our statistics and sequence converge chapters. For now, what does this tell us? The law of large numbers provides insight into the central idea of a limit: what value something approaches as an input or process continues.

5.2 Limits of Asymptotes

A function has a **vertical asymptote** at $x = a$ when the function values grow without bound as x approaches a. A function's **end behavior** describes how the function values behave as $x \to \infty$ or $x \to -\infty$; if the function approaches a fixed value L, we say it has a **horizontal asymptote** at $y = L$.

The asymptotic behavior we see in rational functions suggests that we need to expand our vocabulary of function characteristics. We examined vertical asymptotes and end behavior through graphs and tables, and discussed them in English. The language of limits enables us to discuss these attributes mathematically and with greater efficiency.

Let's revisit an example from the previous chapter. This function has a hole at $x = 1$, a vertical asymptote at $x = 3$, and a horizontal asymptote of $y = 1$.

$$f(x) = \frac{x^2 - 3x + 2}{x^2 - 4x + 3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$$

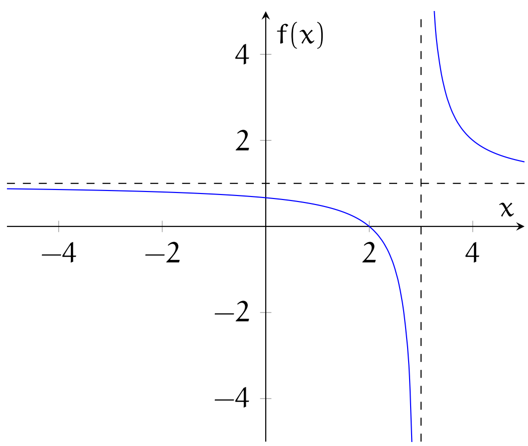


Figure 5.1: Graph of $f(x) = \frac{x^2-3x+2}{x^2-4x+3}$ with asymptotes

First, consider the vertical asymptote. We see that the graph goes down as it hugs the left side of the vertical asymptote, and goes up as it hugs the right side. We can describe these behaviors as the left- and right-hand limits, respectively. We say that the left-hand limit of f at $x = 3$ is negative infinity. Another way of communicating this is to say that as x approaches 3 from the left, the function approaches negative infinity. Symbolically, we summarize this as

$$\lim_{x \to 3^-} f(x) = -\infty$$

The little negative sign in $x \to 3^-$ indicates we are approaching $x = 3$ from the left (the negative side of the axis).

Similarly, the right-hand limit of f at $x = 3$ is positive infinity. In other words, as x approaches 3 from the right, the function approaches positive infinity. Symbolically, we write

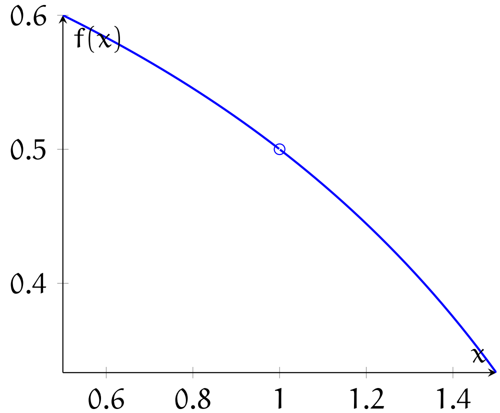
$$\lim_{x \to 3^+} f(x) = \infty$$

This time, the little $^+$ indicates we are approaching the x-value from the right (positive) side of the axis.

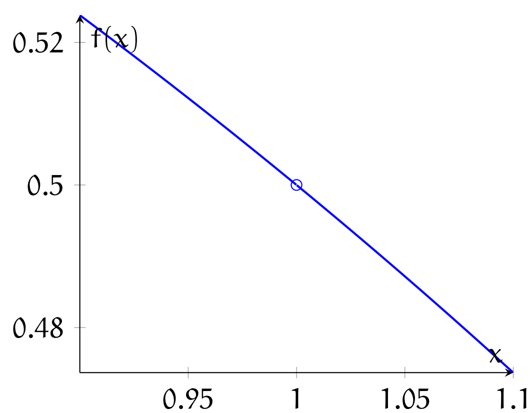
The limit of a function at a particular x-value is the y-value that the function approaches as it approaches the given x-value. In the previous example, we could only specify the left- and right-hand limits, because they were different. In cases where the left- and right-hand limits are equal, we can say that the function has a limit there. The hole in our function f is one such value. We see that as we approach the hole from both the left and right, the function takes on values near $\frac{1}{2}$. This is more apparent numerically:

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	0.5238	0.5025	0.5003	undefined	0.4998	0.4975	0.4737

We can also see this by zooming in on the graph (see Figure 5.2):



(a) Zoomed out near $x = 1$



(b) Zoomed in closer to $x = 1$

Figure 5.2: Two graphs of $f(x) = \frac{x^2-3x+2}{x^2-4x+3}$ zoomed in about $x = 1$

The left-hand and right-hand limits of f at 1 are both $\frac{1}{2}$. Since they are equal, we can also say that the limit of f at 1 is $\frac{1}{2}$. This allows us to efficiently discuss the behavior of f at 1, even though the function is not defined there, as substituting 1 into the function gives division by zero.

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = \lim_{x \to 1} f(x) = \frac{1}{2}$$

We can also talk about limits at x-values where nothing weird is happening (that is, no hole or vertical asymptote). For example, as x approaches 4 from the left and right, y approaches 2.

x	3.9	3.99	3.999	4	4.001	4.01	4.1
$f(x)$	2.1111	2.0101	2.0010	2	1.9990	1.9901	1.9091

In this case, since nothing weird is happening, the limit is equal to the function value. This is an example of continuity, which we will discuss in more detail in the next chapter. By contrast, at the vertical asymptote $x = 1$, since the left- and right-hand limits are not equal, we say the function does not have a limit, or the limit does not exist.

Finally, let's consider the horizontal asymptote of f. The graph hugs the line $y = 1$ as x goes far to the left and far to the right. We say that as x approaches negative infinity, f approaches 1; likewise, that as x approaches positive infinity, f approaches 1. We write these symbolically as $\lim_{x\to-\infty} f(x) = 1$ and $\lim_{x\to\infty} f(x) = 1$.

Exercise 6 Limits Practice 1

Determine the left- and right-hand limits of the function as x approaches the given values. At x-values where the limit exists, determine it.

1. $p(x) = \frac{x+3}{x^2+9x+18}, x = -6, -5, -3, \infty$

Working Space

Answer on Page 114

We have seen two weird behaviors of rational functions at certain x-values: holes and vertical asymptotes. Now, we will examine another type of weird behavior: jumps. This is a characteristic of some piecewise-defined functions. In piecewise-defined functions, the domain is divided into two or more pieces, and a different expression is used to give the y-value depending on which piece contains the x- value. One common piecewise-defined function is the floor function (shown in figure 5.3), sometimes denoted $\lfloor x \rfloor$. The standard floor function rounds any real number down to the nearest integer. So, for a price quoted in dollars and cents, the floor would just be the number of dollars.

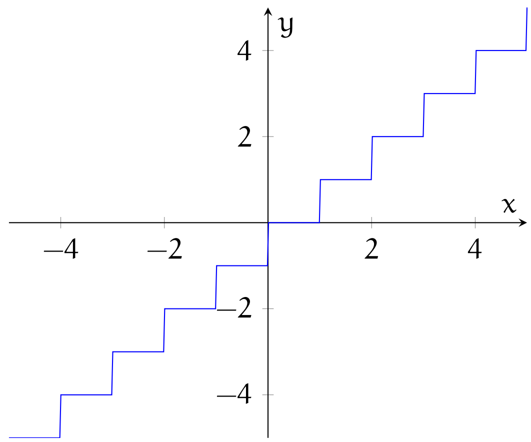


Figure 5.3: Graph of $y = \lfloor x \rfloor$

When x is exactly 1, the function value is 1: the number of dollars in a price of \$1.00. When x is any number greater than 1 but less than 2, the function value is still 1. Also, $\lfloor 1.01 \rfloor$, $\lfloor 1.5 \rfloor$, and $\lfloor 1.99999 \rfloor$ are all 1. As we continue to look to the right, once x equals exactly 2, h jumps up to the value 2. So, $\lim_{x \to 2^-} \lfloor x \rfloor = 1$, while $\lim_{x \to 2^+} \lfloor x \rfloor = 2$.

Besides rational and piecewise defined functions, there are other functions with interesting limits. Consider the standard exponential function, $y = e^x$ (shown in figure 5.4).

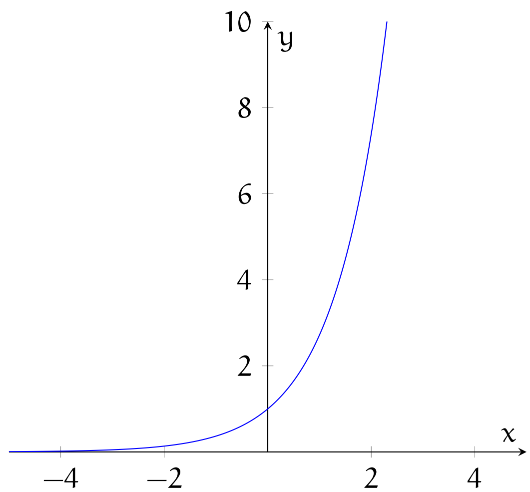


Figure 5.4: Graph of $y = e^x$

As x increases, y increases without bound; that is, $\lim_{x \to \infty} e^x = \infty$. However, looking far to the left, we see that y hugs the x-axis. This is because raising e to a large negative exponent is the same as 1 divided by e raised to a large positive exponent; that is, 1 divided by a very large number, which yields a very small positive number. In limit notation, $\lim_{x \to -\infty} e^x = 0$. This example illustrates that horizontal asymptotes need not model end behavior in both directions. Note that this reasoning holds for $y = b^x$ for any $b > 1$, so all such functions have the same horizontal asymptote, $y = 0$.

We know that the natural logarithm function, $y = \ln x$, is the inverse of $y = e^x$. Since inverse functions swap the role of x and y, it stands to reason that a horizontal asymptote in one function corresponds with a vertical asymptote in the other function, and that is indeed the case (see figure 5.5).

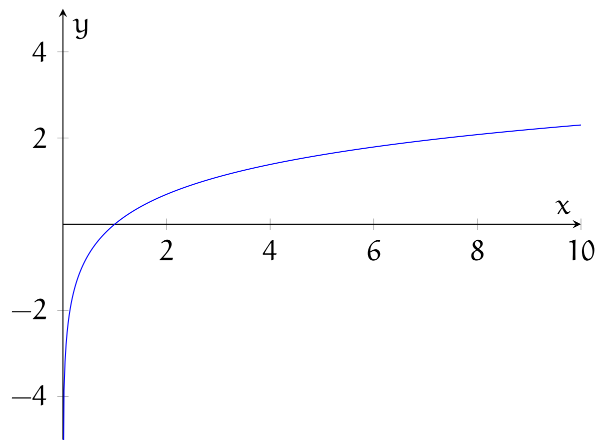


Figure 5.5: Graph of $y = \ln x$

An untransformed logarithm function is defined only for positive inputs. That is because it is not possible to find an exponent of a positive number that will yield a negative or zero result. What type of exponent on a positive number yields a number near zero? That would be a large-magnitude negative number. So, on the logarithm graph, large negative y-values correspond with x-values only slightly greater than zero. So, $\ln x$ (and $\log_2 x$, and indeed $\log_b x$ for any $b > 1$) approaches negative infinity as x approaches 0 from the right. There is no left-hand limit at 0, however. In limit notation, $\lim_{x \to 0^+} \ln x = -\infty$.

Exercise 7 Limits Practice 2

State the asymptotes of the following transformed exponential and logarithmic functions. Give the limit statement which describes the behavior of the function along the asymptote.

1. $y = 3^x + 1$
2. $y = \log_2(x - 4)$
3. $y = 2^{1-x}$
4. $y = \log_{10}(-2x)$

Working Space

Answer on Page 114

We next consider two functions that each have two horizontal asymptotes. These two seemingly obscure functions are quite important in data science.

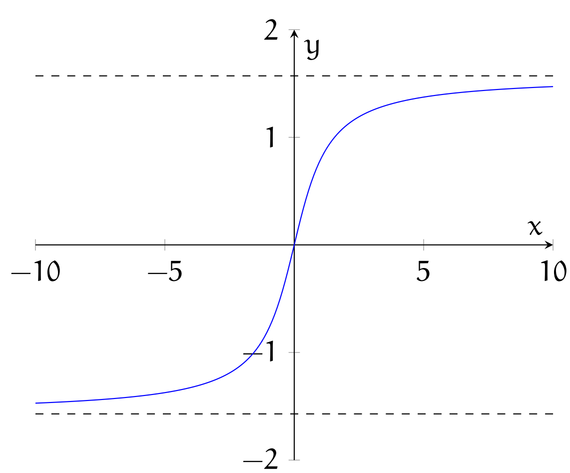


Figure 5.6: Graph of $y = \arctan x$

We know that the arctangent, or inverse tangent, function is the inverse of the piece of the tangent function which passes through the origin. The vertical asymptotes bounding this piece become horizontal asymptotes when the function is inverted.

Here are the equation and graph of the logistic function:

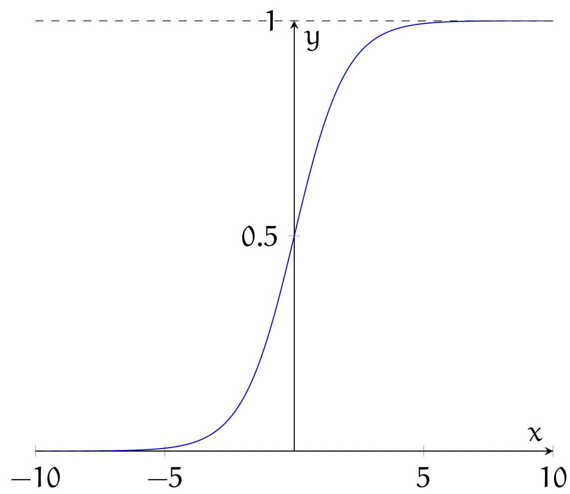


Figure 5.7: Graph of the logistic function, $y = \frac{1}{1+e^{-x}}$

For large magnitude negative values of x, the exponential term in the denominator becomes a very large positive value. The fraction thus becomes a positive number very close to zero. For large magnitude positive values of x, that exponential term becomes a very small positive number. Adding it to 1 yields a denominator just barely greater than 1. Dividing 1 by this number therefore yields a function value just barely less than 1. So, the logistic function yields values between 0 and 1, though never equaling either of these values exactly. It is precisely this characteristic which makes the logistic function so useful.

Exercise 8 Limits Practice 3

Using limit notation, state the limits as x approaches negative and positive infinity for the inverse tangent and logistic functions given above.

Working Space

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As seen above, the limit of a function from the left may be different from the limit of the function from the right. Additionally, the actual *value* of the function may be different from the limit. Consider the piecewise function $h(x)$:

$$h(x) = \begin{cases} -x^2 + 3, & \text{if } x < 0 \\ 2, x = 0 \\ -x + 3, & \text{if } x > 0 \end{cases}$$

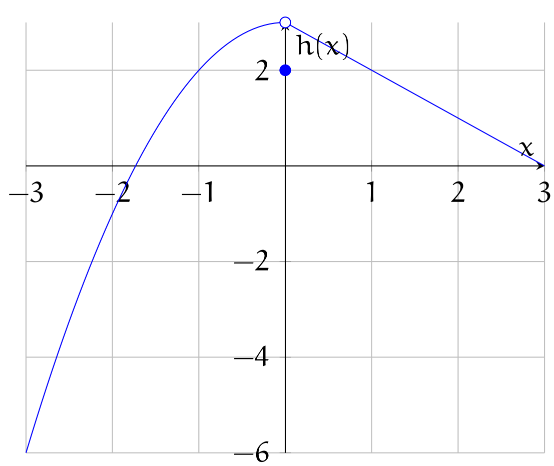


Figure 5.8: Graph of the piecewise function, $h(x)$

From examining the graph, we see that

$$\lim_{x \to 0_-} h(x) = \lim_{x \to 0_+} h(x) = 3$$

However, $h(0) = 2 \neq 3$. So, does this limit exist? It does! The limit of a function describes the *behavior* of the function around a particular value, not the value of the function itself. In order for a limit to exist, the limits from the left and right must be equal to each other, but not necessarily the actual value of the function.

Use the graph of $h(x)$ above and the graphs of $f(x)$ and $g(x)$ below to complete the following exercise.

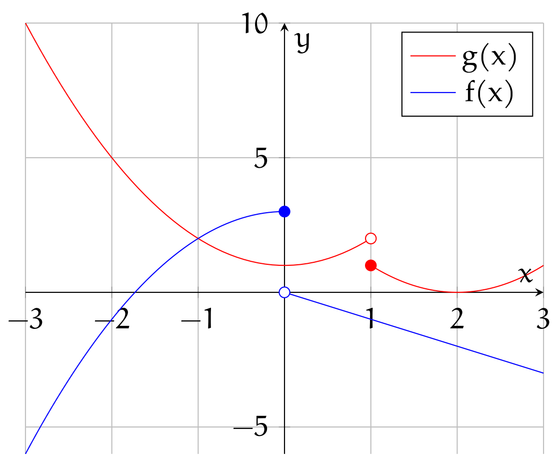


Figure 5.9: Piecewise functions $f(x)$ and $g(x)$

Exercise 9 Limits Practice 4

Determine the limit from the left and the right for each function at the given value(s). State the limit at that value, if it exists.

1. $h(x), x = -1, 0, 1$
2. $f(x), x = -1, 0, 2$
3. $g(x), x = -2, 0, 1, 2$

Working Space

Answer on Page 114

5.3 Continuity

A note about continuity:

In order to be able to talk more about limits and know when we can apply certain rules and theorems, we first must discuss continuity. A function is continuous if there are no ″jumps″ or ″gaps″ in the graph of the function. For example, the function $f(x) = x^2$ is continuous for all real values of x. On the other hand, the function $g(x) = \tan(x)$ has many discontinuities, including at $x = \frac{\pi}{2}$. Let's examine the graph of each of these functions:

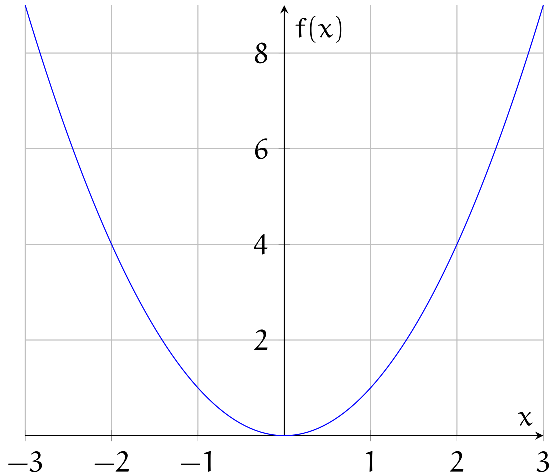


Figure 5.10: Graph of $f(x) = x^2$

If you wanted, you could trace your finger along the graph of $f(x)$ from $x = -3$ to $x = 3$ without ever picking up your finger. This means the function is continuous in the domain from $-3 \leq x \leq 3$. In this case, the domain of continuity *includes* the end points ($x = 3$ and $x = -3$). This is called a closed interval. In other cases, the function will be continuous right up to, but not including, the endpoints, as with the domains of continuity for our other example, $g(x) = \tan x$. This is called an open interval. Let's learn more about intervals of continuity by examining $g(x) = \tan x$.

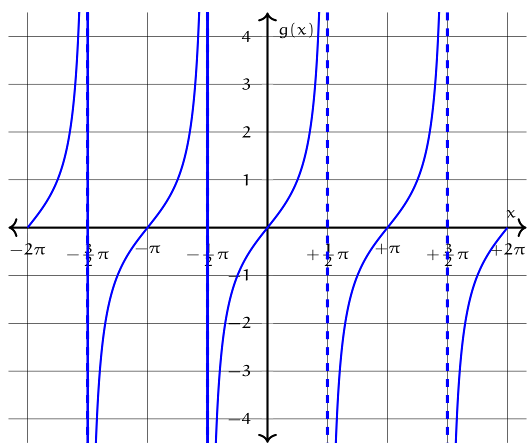


Figure 5.11: Graph of $g(x) = \tan x$

As you can see, if you trace your finger along the graph of the function starting at $x = 0$, you can continue without lifting your finger to $x = \frac{\pi}{2}$. As you approach $x = \frac{\pi}{2}$ from the left, the value of $g(x)$ approaches ∞. In order to continue tracing the function PAST $x = \frac{\pi}{2}$, you have to lift your finger and bring it down to $-\infty$. The function then continues continuously again until $x = \frac{3\pi}{2}$.

In the case of $g(x) = \tan x$, the function is continuous on *open intervals*, including the open interval $\frac{\pi}{2} < x < \frac{3\pi}{2}$.

There is a shorter way to represent open and closed domain intervals. We can represent that $f(x) = x^2$ is continuous on the closed interval $-3 \leq x \leq 3$ in the following way:

$$x \in [3, -3]$$

This reads as "x contained in the domain -3 to 3, inclusive". That is, all the values from -3 to 3, including the endpoints. The inclusion of the endpoints is implied by the use of *brackets*. For open intervals, we use parentheses to communicate that the interval goes up to, but does not include, the endpoints. For $g(x) = \tan x$, we can use parentheses:

$$x \in \left(\frac{-3\pi}{2}, \frac{-\pi}{2}\right)$$

because the $g(x) = \tan x$ is not continuous at $x = \frac{-3\pi}{2}$ or at $x = \frac{-\pi}{2}$.

Formally, a function $f(x)$ is continuous at $x = a$ if $\lim_{x \to a} f(x)$ exists *and* $\lim_{x \to a} f(x) = f(a)$. In other words, the limit is equal to the actual value of the function. Re-examine the graph of $h(x)$. We have already seen that $\lim_{x \to 0} h(x)$ exists and is equal to 3. However, $h(0) = 2 \neq \lim_{x \to 0} h(x)$. So $h(x)$ is not continuous at $x = 0$. Because $-x^2+3$ is evaluable all the way to $-\infty$ and $-x+3$ is evaluable all the way to ∞, the function $h(x)$ is continuous everywhere *except* $x = 0$. We can represent this mathematically by saying $h(x)$ is continuous on the domain $x \in (-\infty, 0) \cup (0, \infty)$. We use parentheses for $\pm\infty$ because we can never actually reach ∞. Additionally, the function is continuous up to, but not including 0, and the use of parentheses excludes $x = 0$ from the domain of continuity.

5.3.1 Continuity Practice

Exercise 10

[This problem was originally presented as a calculator-allowed, multiple- choice question on the 2012 AP Calculus BC exam.] Suppose a function f is continuous at $x = 3$. Classify the following statements as always true, sometimes true, or never true. Explain your answers.

1. $f(3) < \lim_{x \to 3} f(x)$
2. $\lim x \to 3^+ f(x) \neq \lim_{x \to 3^-} f(x)$
3. $f(3) = \lim_{x \to 3^+} f(x) = \lim_{x \to 3^-} f(x)$
4. The derivative of f at $x = 3$ exists.
5. The derivative of f is positive for $x < 3$ and negative for $x > 3$.

Working Space

Answer on *Page 115*

Exercise 11 Limits Practice 5

State the location of discontinuities (if any) and explain why the function is discontinuous at that location:

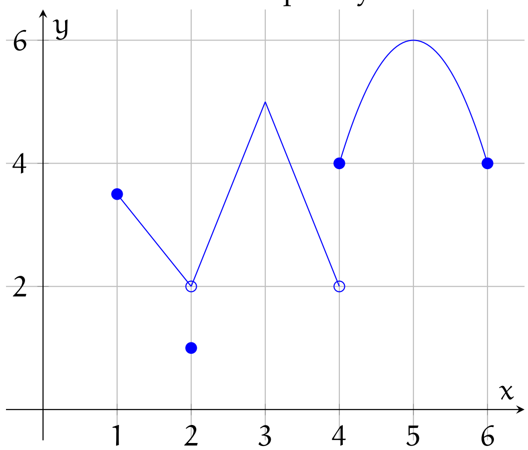
1. $f(x) = \frac{3x^2 - 8x - 3}{x - 3}$
2. $f(x) = \begin{cases} \frac{2}{x^4}, & \text{if } x <\neq 0 \\ 2, & \text{if } x = 0 \end{cases}$
3. $f(x) = \begin{cases} \frac{3x^2 - 8x - 3}{x - 3}, & \text{if } x \neq 3 \\ 1, & \text{if } , x = 3 \end{cases}$

Working Space

Answer on *Page 115*

Exercise 12

The graph of a function, $h(x)$, is shown. Classify each of the following statements as true or false and explain your answer.



1. $\lim_{x \to 2} h(x)$ exists
2. $\lim_{x \to 3} h(x)$ does not exist
3. $\lim_{x \to 4} h(x)$ exists
4. $h(x)$ is continuous at $x = 5$
5. $h(x)$ is not continuous at $x = 4$
6. $\lim_{x \to 2} h(x) = h(2)$

Working Space

Answer on Page 116

5.4 Limits Rules

There are some mathematical properties of limits which allow us to determine the limit of complex functions without seeing a graph or using a calculator to generate a table.

The following laws are true given that c is a constant, $\lim_{x \to a} f(x)$ exists, and $\lim_{x \to a} g(x)$ exists.

1. Sum Law $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$

2. Difference Law $\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$
3. Constant Multiple Law $\lim_{x\to a}[cf(x)] = c \cdot \lim_{x\to a} f(x)$
4. Product Law $\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$
5. Quotient Law $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ given that $\lim_{x\to a} g(x) \neq 0$

These laws are fairly obvious — the limit of the sum of two functions is equal to the sum of the limits of each function individually. The only tricky one is the last: The limit of the quotient of two functions is equal to the quotient of the limits if and only if the limit of the function in the denominator does not equal zero. This makes sense, as we know dividing by zero yields an undefined result.

Let's practice applying these laws to evaluate the limits of the functions $f(x)$, shown in blue below, and $g(x)$, shown in red below:

$$f(x) = \begin{cases} -x^2+3, & \text{if } x \leq 0 \\ -x, & \text{if } x > 0 \end{cases}$$

$$g(x) = \begin{cases} x^2+1, & \text{if } x < 1 \\ (x-2)^2, & \text{if } x \geq 1 \end{cases}$$

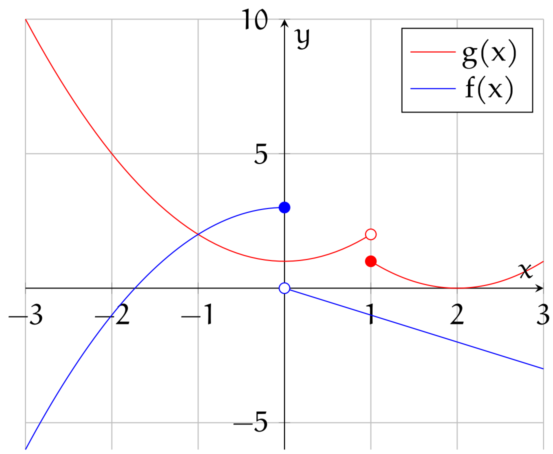


Figure 5.12: Graphs of the piecewise functions $f(x)$ and $g(x)$

We can use these laws to evaluate limits involving $f(x)$ and $g(x)$ (shown on the graph above). Here are some examples: Use the graphs of $f(x)$ and $g(x)$ given above to evaluate each limit, if it exists. If the limit does not exist, explain why. Two examples are given first:

Example 1: Evaluate $\lim_{x\to 0} f(x) \cdot g(x)$

Solution 1: From the Product Law, we know that:

$$\lim_{x \to 0} f(x) \cdot g(x) = \lim_{x \to 0} f(x) \cdot \lim_{x \to 0} g(x)$$

Looking at the graph, we can see that

$$\lim_{x \to 0} g(x) = 1$$

and there is a discontinuity in $f(x)$ at $x = 0$. Therefore,

$$\lim_{x \to 0} f(x) = \text{DNE}$$

Substituting this, we get:

$$\lim_{x \to 0} f(x) \cdot \lim_{x \to 0} g(x) = \text{DNE} \cdot 1 = \text{DNE}$$

Therefore, *the limit does not exist*.

Example 2: Evaluate $\lim_{x \to 2} f(x) - g(x)$

Solution 2: Applying the Difference Law, we see that:

$$\lim_{x \to 2} [f(x) - g(x)] = \lim_{x \to 2} f(x) - \lim_{x \to 2} g(x)$$

Examining the graph, we see that

$$\lim_{x \to 2} f(x) = -2$$

and

$$\lim_{x \to 2} g(x) = 0$$

Substituting these values, we get:

$$\lim_{x \to 2} f(x) - g(x) = -2 - 0 = -2$$

Exercise 13 Limits Practice 6

Working Space

1. $\lim_{x\to -3} \frac{f(x)}{g(x)}$
2. $\lim_{x\to 2} [f(x) + 5g(x)]$
3. $\lim_{x\to -1} \frac{3g(x)}{f(x)}$
4. $\lim_{x\to 0} f(x) \cdot 5g(x)$
5. $\lim_{x\to -1} f(x) - 3g(x)$

Answer on Page 116

Recall that exponents represent repeated multiplication. Therefore, if we apply the Product Law multiple times, we obtain the Power Law for limits:

6. Power Law $\lim_{x\to\infty} [f(x)]^n = [\lim_{x\to\infty} f(x)]^n$ where n is a positive integer

There are two special limits that will be useful to us and are intuitively obvious, but we won't formally prove here.

7. $\lim_{x\to a} c = c$
8. $\lim_{x\to a} x = a$

Combining Law 8 with the Power Law, we find that:

9. $\lim_{x\to a} x^n = a^n$

And similarly, for square roots:

10. $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$ (if n is even, we assume $a > 0$)

Direct substitution property: If f is a polynomial or rational function and a is in the domain for f, then

$$\lim_{x\to a} f(x) = f(a)$$

Often, rational functions can be simplified. In an above example, we computed the limit by simplifying $f(x) = \frac{3x^2-8x-3}{x-3}$ to the simpler $g(x) = 3x+1$. This is a valid strategy because $\frac{3x^2-8x-3}{x-3} = 3x+1$ when $x \neq 3$. Remember: a limit describes how a function behaves *as it approaches* a, not its value/behavior when x *actually equals* a. This reveals the following useful rule:

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$, provided the limit exists.

5.5 Squeeze Theorem

The Squeeze Theorem states that if $f(x) \leq g(x) \leq h(x)$ when x is near a (except at a) and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$

In other words, if $g(x)$ is between $f(x)$ and $h(x)$ near a, and f and h have the same limit, L, then the limit of g must also be L.

Example: Let's examine the graph of $g(x) = x^2 \sin\frac{1}{x}$ and determine $\lim_{x\to 0} g(x)$:

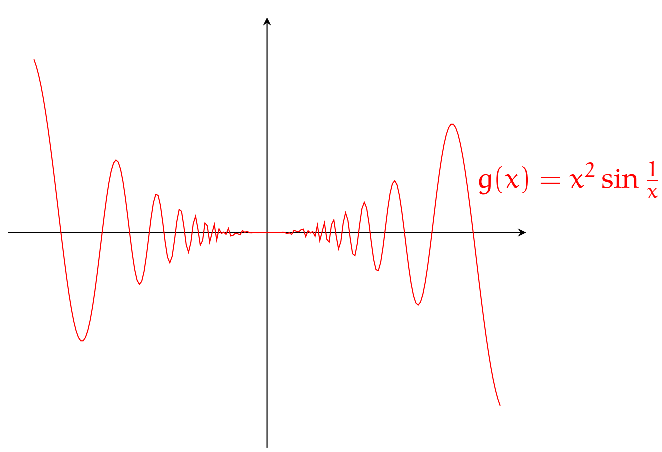


Figure 5.13: Graph of $g(x) = x^2 \sin\frac{1}{x}$

Solution: Because $\sin\frac{1}{x}$ is undefined at $x = 0$, we cannot compute the limit directly. However, from examining the graph, we can guess that $\lim_{x\to 0} g(x) = 0$. Feel free to confirm this with your calculator. We need to choose two functions: one that is larger than $g(x)$ near $x = 0$ and one that is smaller. Since $|\sin\frac{1}{x}| \leq 1$ (when $x \neq 0$), then

$$|x^2 \sin\frac{1}{x}| \leq x^2$$

and

$$-x^2 \leq x^2 \sin\frac{1}{x} \leq x^2$$

Let's confirm this by plotting $f(x) = -x^2$, $g(x)$, and $h(x) = x^2$ on the same graph:

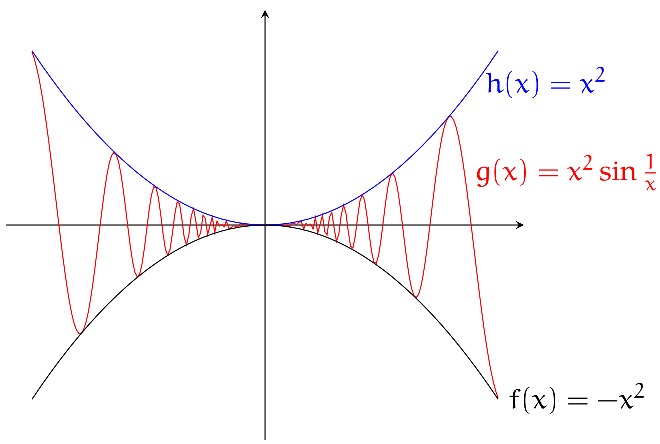


Figure 5.14: Squeeze Theorem example

As you can see, when x is near 0, $f(x) \leq g(x) \leq h(x)$. Because $f(x)$ and $h(x)$ are both polynomials, their limits are straightforward:

$$\lim_{x\to 0} -x^2 = 0 \text{ and } \lim_{x\to 0} x^2 = 0$$

Then, by the Squeeze Theorem, we can say that:

$$\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$$

5.5.1 Squeeze Theorem Practice

Exercise 14 Squeeze Theorem 1

Use the Squeeze Theorem to show that $\lim_{x \to 0} \sqrt{x^3 + x^2} \cos \frac{1}{x} = 0$. Illustrate by graphing the functions you define as f, g, and h on the same plot.

Working Space

Answer on Page 118

Exercise 15 Squeeze Theorem 2

If $2x+3 \leq f(x) \leq x^2-2x+7$ for $x \geq 0$, find $\lim_{x\to 2} f(x)$

Working Space

Answer on Page 118

Exercise 16 Squeeze Theorem 3

Prove that $\lim_{x\to 0^+} \sqrt{x}e^{\sin\frac{\pi}{x}} = 0$

Working Space

Answer on Page 119

5.6 Intermediate Value Theorem

When considering functions that are continuous on a closed interval, the Intermediate Value Theorem can help us: Given a function, $f(x)$, that is continuous on the closed interval $[a,b]$ and $f(a) \neq f(b)$, there is at least one number c such that $f(c) = N$, where N is any number between $f(a)$ and $f(b)$. The theorem is illustrated in figures 5.15 and 5.16:

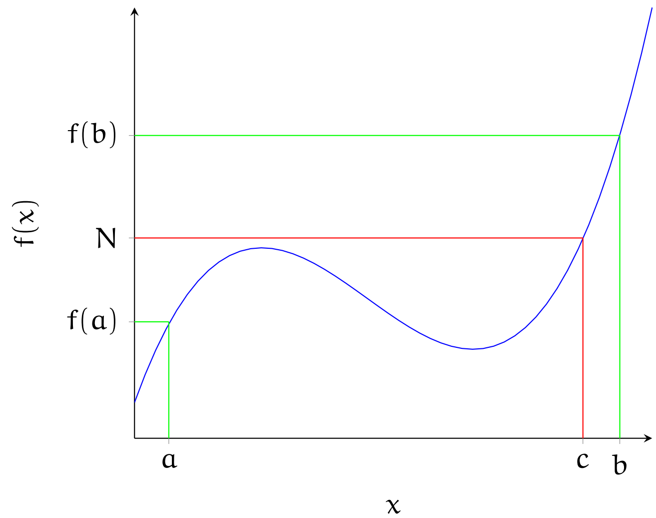


Figure 5.15: An example where one solution satisfies the Intermediate Value Theorem

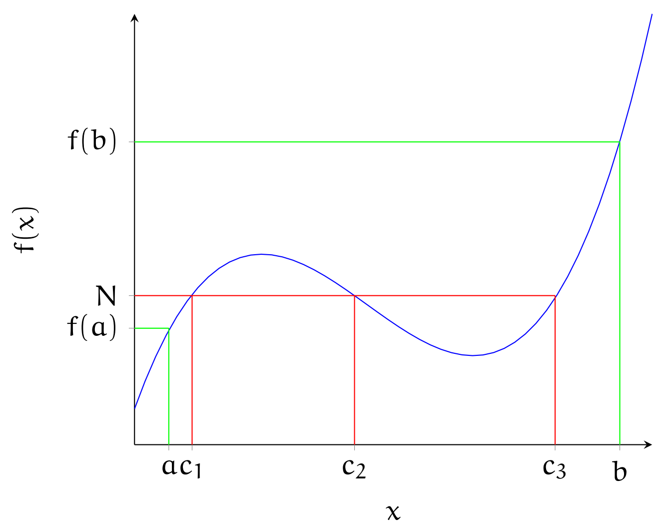


Figure 5.16: An example where more than one solution satisfies the Intermediate Value Theorem

Logically, we can think of the IVT this way: If a function is continuous on a closed interval, there are no gaps or breaks. If there are no gaps or breaks, then the function must pass through the line $y = N$, since it cannot jump over the line. Your graphing calculator uses IVT to find roots of functions:

Example: Show that there is at least one root of the equation $2x^3 - 6x^2 + 3x - 1 = 0$ between $x = 2$ and $x = 3$.

Solution: For IVT to apply, we must first check that the function is continuous on the closed interval $x \in [2, 3]$. We define $f(x) = 2x^2 - 6x^2 + 3x - 1$, which is continuous everywhere, because it is a polynomial function. *For more complex functions, always be sure to check the endpoints of an interval, since IVT only applies on closed intervals of continuity*. We will take $a = 2$, $b = 3$, and $N = 0$. We find the values of $f(x)$ at the endpoints:

$$f(2) = 2(2)^3 - 6(2)^2 + 3(2) - 1 = 16 - 24 + 6 - 1 = -3$$
$$f(3) = 2(3)^3 - 6(3)^2 + 3(3) - 1 = 54 - 54 + 9 - 1 = 8$$

Therefore, $f(2) < 0 < f(3)$ and according to IVT, there must exist some c such that $f(c) = 0$ and there is a root to the equation in the interval $x \in (2, 3)$.

Exercise 17 Intermediate Value Theorem Practice

Use the IVT to show there is a solution the given equation on the stated interval:

1. $2x^4 + x - 12 = 0\ ,\ (1, 2)$
2. $\ln(x) = 3x - 4\sqrt{x}\ ,\ (2, 3)$
3. $2\sin x = 3x^2 - 2x\ ,\ (1, 2)$

Working Space

Answer on Page 119

Chapter 6

Parametric Functions

Throughout your study of functions, you have seen functions that look like this:

$$y = f(x)$$

Sometimes, to describe certain shapes or curves on a graph, we must use more than one function to do so. This means a shape, like a circle on the graph, can be represented using

$$y = f(t) \quad \text{and} \quad x = g(t).$$

In this case, there are two functions, $x(t)$ and $y(t)$, that depend on a third variable t, and together they help us visualize and define a curve.

The central word you should think about as you go through this chapter is *parameter*. A parameter is a variable that both x and y depend on. The most common example of a parametric function is the circle. To represent a circle on a graph, we can use the equations

$$x = \cos(\theta) \quad \text{and} \quad y = \sin(\theta).$$

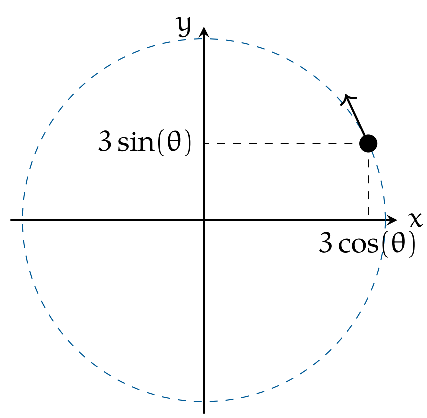


Figure 6.1: A diagram of a circle formed through a parametric functions.

In this case, the graph of the circle is determined by the value of the angle θ, the parameter.

So, why parametric functions? When you look up the function that describes a circle, you might see something like

$$x^2 + y^2 = r^2.$$

The equation $x^2+y^2 = r^2$ does not describe y as a single function of x. You would need two functions (one for the top half and one for the bottom half), and many curves cannot be written as a single function at all. Parametric equations solve this problem by allowing us to describe the entire curve with one pair of equations, using a parameter t. This parameter can be denoted by other letters, but we will use t in this chapter.

Imagine a ball rolling across graph paper. At each moment, the ball is at a different location. If we label time with a variable t, then for every value of t the ball has an x-coordinate and a y-coordinate. What if we had two equations, one that told us its x-position at time t and one that told us its y-position? As t changes, these two equations would trace the entire path of the ball. This is the basic idea behind parametric equations.

6.1 Conversion to Rectangular Form

In many cases, we can eliminate the parameter to convert a pair of parametric equations,

$$x = f(t), \quad y = g(t),$$

into a single equation relating x and y.

In many cases, a parametric curve can be rewritten as a single equation involving only x and y. This process is called eliminating the parameter, and the resulting equation is known as the rectangular form of the curve.

To eliminate the parameter, we solve one of the parametric equations for t and substitute into the other equation.

Example:

Consider the parametric equations

$$x = 2t + 1, \quad y = 3t - 4.$$

Solution: Solve the equation for x to isolate t:

$$t = \frac{x-1}{2}.$$

Substitute into the equation for y:

$$y = 3\left(\frac{x-1}{2}\right) - 4.$$

Simplifying,

$$y = \frac{3}{2}x - \frac{11}{2}.$$

Example

Consider the parametric equations

$$x = \cos t, \quad y = \sin t.$$

Solution: In this case, solving for t directly is not convenient. Instead, we use the trigonometric identity

$$\cos^2 t + \sin^2 t = 1.$$

Substituting $x = \cos t$ and $y = \sin t$ gives

$$x^2 + y^2 = 1.$$

This rectangular equation describes a circle of radius 1 centered at the origin.

Note

Converting to rectangular form describes the shape of a curve, but it does not show the direction of motion or how many times the curve is traced. This information is contained in the parametric equations and the domain of the parameter!

6.2 Sketching Parametric Equations

Parametric equations describe curves by expressing both x and y in terms of a third variable, usually t, called the parameter. When sketching a parametric curve, it is important to remember that the curve is traced as the parameter changes, rather than being drawn all at once as a function of x.

To sketch a parametric curve,

- choose several values of the parameter t,
- compute the corresponding x- and y-values,
- plot the resulting points in the xy-plane,
- and indicate the direction in which the curve is traced as t increases.

Unlike rectangular graphs, parametric sketches also depend on the direction and speed of motion, which are determined by how $x(t)$ and $y(t)$ change as t varies.

FIXME Sketched example

Exercise 18

Sketch the parametric equations

$$x = t + 1, \quad y = t^2.$$

In your sketch:

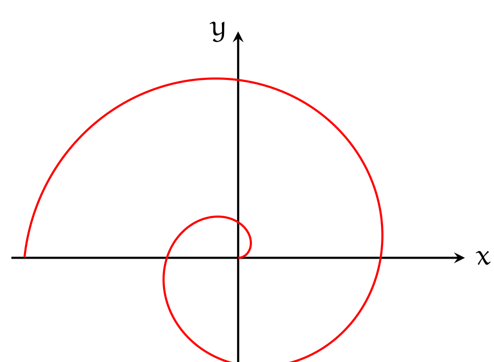
- create a table of values for t, x, and y,
- plot the corresponding points,
- and indicate the direction of motion as t increases.

Working Space

Answer on Page 120

Here is an example of an Archimedean Spiral, a spiral where the radius r is proportional to the angle θ. In parametric form, it is represented as

$$x(t) = t\cos(t), \quad y(t) = t\sin(t)$$



6.3 Using Python to Explore Parametric Curves

Python can be used to visualize parametric curves and explore how the point $(x(t), y(t))$ moves as t changes. For example, we could write code that:

- plots points along a circle defined by $x = \cos(t)$ and $y = \sin(t)$,
- shows the distance from the origin,
- traces more complicated parametric curves like spirals or ellipses.

Below is a Python script that creates an animation for sketching parametric curves. There is a provision for changing the equations based on exercises and questions you come across.

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

t = np.linspace(0, 3, 50)

x = 4 - 2*t
y = 3 + 6*t - 4*t**2

fig, ax = plt.subplots()
ax.set_aspect("equal", "box")

ax.axhline(0, color="black", linewidth=0.5)
ax.axvline(0, color="black", linewidth=0.5)

ax.set_xlabel("x(t)")
ax.set_ylabel("y(t)")
ax.set_title("Parametric Equation of Curve")

ax.set_xlim([x.min() - 1, x.max() + 1])
ax.set_ylim([y.min() - 1, y.max() + 1])

(line,) = ax.plot([], [], "b--")
(point,) = ax.plot([], [], "ro")

def update(frame):
    line.set_data(x[:frame + 1], y[:frame + 1])
    point.set_data([x[frame]], [y[frame]])
    return line, point
```

```

ani = FuncAnimation(fig, update, frames=len(t), interval=50, blit=True)
ani.save("parametric_animation.mp4")

plt.show()
```

Exercise 19

Working Space

Consider the parametric equations

$$x = 4 - 2t, \quad y = 3 + 6t - 4t^2,$$

for $0 \leq t \leq 3$.

(a) Sketch the curve by creating a table of values for t, x, and y.

(b) Use the provided Python animation code to visualize the curve.

(c) Describe the direction in which the curve is traced as t increases.

(d) Explain how the Python animation supports or clarifies your hand-drawn sketch.

Answer on Page 120

6.4 Derivatives of Parametric Functions

Just like any other type of curve, you can be asked to determine the slope of the tangent to the point on a parametric curve. To do this, you should know how to find the derivative of the curve! You could do this by eliminating the variable t and solving for the derivative of the rectangular equation. However, there is an easier way!

The derivative of a parametric curve can be found by dividing $\frac{dy}{dt}$ by $\frac{dx}{dt}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

Derivatives of parametric functions are used to describe the slope of a curve at a given value of the parameter. This slope indicates how the curve is changing at that point. In some problems, the slope is used to find the equation of a tangent line to the curve. In other problems, the slope is used to analyze the behavior of the curve, such as determining where the curve has horizontal or vertical tangents.

Example

Find $\dfrac{dy}{dx}$ for the parametric equations

$$x = t^2 + 1, \quad y = 3t - 2.$$

Solution

First compute the derivatives with respect to t:

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3.$$

Now divide:

$$\frac{dy}{dx} = \frac{3}{2t}.$$

As a check, eliminate the parameter. From $x = t^2 + 1$, we get

$$t = \sqrt{x - 1}.$$

Substitute into $y = 3t - 2$:

$$y = 3\sqrt{x - 1} - 2.$$

Differentiating gives

$$\frac{dy}{dx} = \frac{3}{2\sqrt{x-1}},$$

which matches the parametric result.

Exercise 20

Working Space

Consider the parametric equations

$$x = t^3 - 3t, \quad y = t^2.$$

(a) Compute $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

(b) Find the values of t for which the curve has horizontal tangents.

(c) Find the values of t for which the curve has vertical tangents.

(d) Identify the points on the curve where these tangents occur.

Answer on Page 121

Exercise 21

Consider the parametric equations

$$x = 2t + 1, \quad y = t^2.$$

(a) Find $\dfrac{dy}{dx}$ in terms of t.

(b) Evaluate $\dfrac{dy}{dx}$ at $t = 1$.

(c) Interpret your result as the slope of the tangent line at the corresponding point.

Working Space

Answer on Page 121

FIXME Expand on section

6.5 Length of Curve Traced by Parametric Equations

In the Python animation, a point moves along the curve as the parameter t increases. This animation highlights an important idea: parametric equations do not only describe the shape of a curve, they describe motion along that curve.

At each moment in time, the point is located at $(x(t), y(t))$. As t changes, the point moves from one position to the next. Sometimes this movement covers a small distance, and sometimes it covers a larger distance, even though the point remains on the same curve. This means the point does not necessarily move at a constant rate.

Speed measures how fast the point is moving along the curve at a given value of t. Once the speed is known, we can determine how far the point travels over a time interval. The length of the curve traced by the parametric equations is the total distance traveled by the moving point.

Speed of a Parametric Curve

When a point moves along a parametric curve, its velocity is determined by the derivatives $x'(t)$ and $y'(t)$. The speed of the point is the magnitude of the velocity vector:

$$\text{Speed} = \sqrt{(x'(t))^2 + (y'(t))^2}.$$

Speed describes how fast the point is moving at a particular moment, but it does not tell us the total distance traveled along the curve.

Arc Length Formula

To find the length of the curve traced by a parametric equation from $t = a$ to $t = b$, we integrate the speed over that interval.

If a curve is defined parametrically by $x = x(t)$ and $y = y(t)$, where $x'(t)$ and $y'(t)$ are continuous on $[a, b]$, then the length of the curve is given by

$$L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\, dt.$$

Example: Finding Speed

Given
$$x'(t) = 8t - t^2, \quad y'(t) = -t + \sqrt{t^{1.2} + 20},$$
the speed of the particle at time t is
$$\text{Speed} = \sqrt{(8t - t^2)^2 + \left(-t + \sqrt{t^{1.2} + 20}\right)^2}.$$

Evaluating at $t = 2$ gives a speed of approximately 12.3 cm/s.

Example: Length of a Curve

Consider the parametric equations
$$x = \cos t, \quad y = \sin t,$$
for $0 \le t \le 2\pi$.

First compute the derivatives:
$$x'(t) = -\sin t, \quad y'(t) = \cos t.$$

Substitute into the arc length formula:
$$L = \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2}\, dt.$$

Using the identity $\sin^2 t + \cos^2 t = 1$, this simplifies to
$$L = \int_0^{2\pi} 1\, dt = 2\pi.$$

Thus, the length of the curve traced is 2π, which matches the circumference of the unit circle.

Note

Speed describes how fast a point moves along a parametric curve at a given moment. Arc length measures the total distance traveled along the curve over an interval of time. For parametric curves, arc length is found by integrating the speed.

Exercise 22

How can two sets of parametric equations represent the same graph, but different curves?

Working Space

Answer on Page 122

Chapter 7

Vector-valued Functions

In the last chapter, you calculated the flight of the shell. For any time t, you could find a vector $[\mathtt{distance}, \mathtt{height}]$. This can be thought of as a function f that takes a number and returns a 2-dimensional vector. We call this a *vector-valued* function from $\mathbb{R} \to \mathbb{R}^2 \to \mathbb{R}^3$[1].

7.1 Vector-valued functions: position

We often make a vector-valued function by defining several real-valued functions. For example, if you threw a hammer with an initial upward speed of 12 m/s and a horizontal speed of 4 m/s along the x axis from the point $(1, 6, 2)$, its position at time t (during its flight) would be given by:

$$f(t) = [4t + 1,\ 6,\ -4.8t^2 + 12t + 2]$$

In other words, x is increasing with t, y is constant, and z is a parabola.

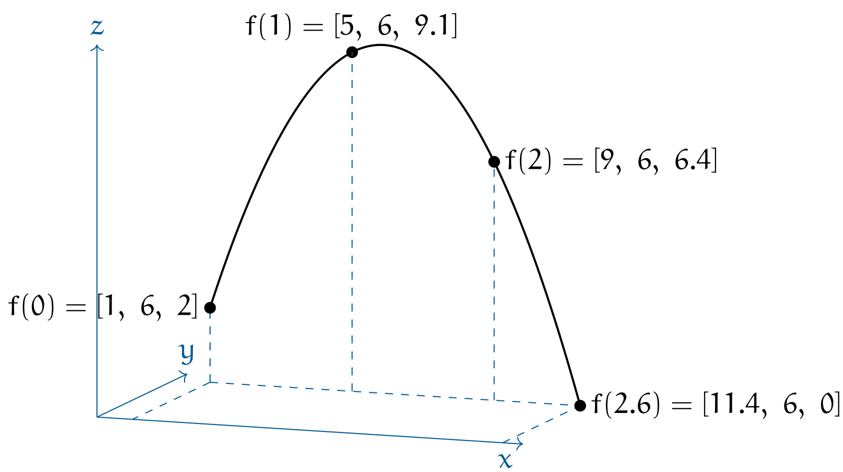


Figure 7.1: An example of a vector-valued function.

[1] the $\mathbb{R}$ symbol represents the set of all real numbers; the $\mathbb{R}^2$ symbol represents the set of all 2-dimensional vectors, and $\mathbb{R}^3$ represents the set of all 3-dimensional vectors

7.2 Finding the velocity vector

Now that we have its position vector, we can differentiate each component separately to get its velocity as a vector-valued function:

$$f'(t) = [4, 0, -9.8t + 12]$$

In other words, the velocity is constant along the x-axis, zero along the y-axis, and decreasing with time along the z axis.

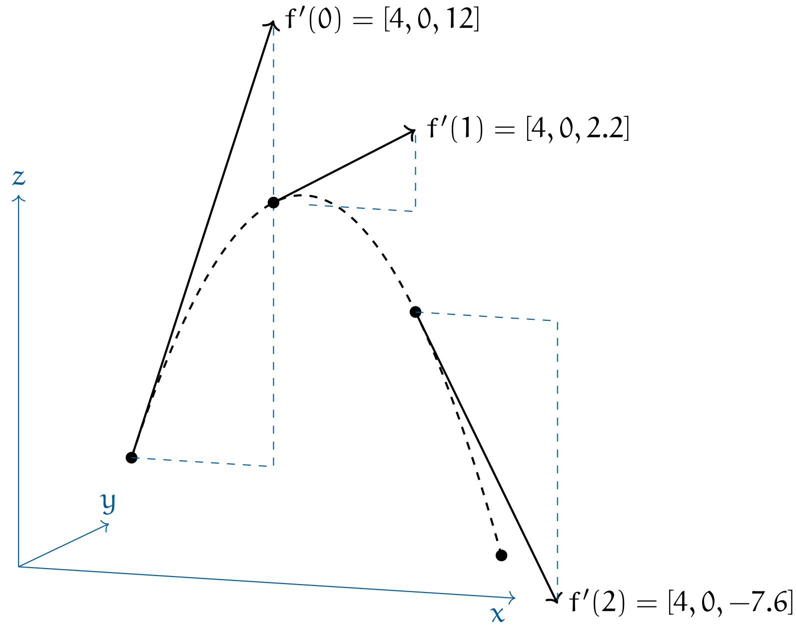


Figure 7.2: The derivatives of the position function (velocity) with respect to time.

7.3 Finding the acceleration vector

Now that we have its velocity, we can get its acceleration as a vector-valued function:

$$f''(t) = [0, 0, -9.8]$$

There is no acceleration along the x or y axes. It is accelerating down at a constant $9.8m/s^2$.

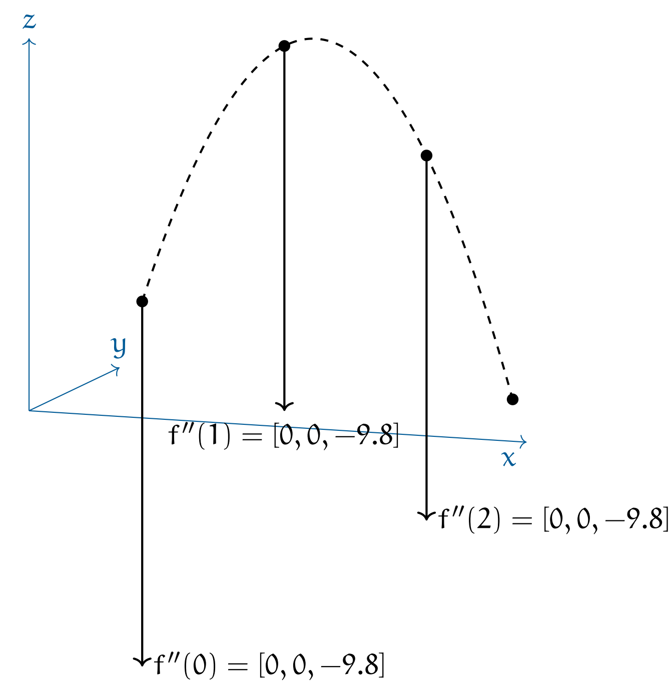


Figure 7.3: The acceleration vector is constant and points downward.

Chapter 8

Circular Motion

We can recall the motion of 2D objects, such as a projectile launched or a car driving along a road. They have components of linear motion, such as position, velocity, acceleration, and force.

But that only works for objects moving in a straight line. What about objects moving in a circle?

8.1 Introduction to Uniform Circular Motion

Recall that in linear motion, You may be surprised to know that we can use very similar equations to describe circular motion, provided you use the correct variables.

The angular displacement, θ, is the angle in *radians* that the object has traveled. The angular velocity, ω, is the rate of change of the angular displacement, and the angular acceleration, α, is the rate of change of the angular velocity.

Linear motion	Rotational motion
$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
$x = x_0 + v_0 t + \frac{1}{2}at^2$	$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

Table 8.1: Kinematic equations for linear and rotational motion.

Linear velocity, v, is related to angular velocity by the equation:

$$v = r\omega$$

The *Period*, T, is the time it takes to complete one full rotation. The frequency, f, is the number of rotations per second. The two are related by:

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

$$f = \frac{1}{T}$$

The *centripetal force*, usually labelled differently (such as tension or gravity), is given by the equation:

$$F = mr\omega^2 = \frac{mv^2}{r}$$

The *angular velocity,* ω is rotations with respect to time. It can be defined in the following ways:

$$\omega_{inst} = \frac{d\theta}{dt}, \qquad \omega_{avg} = \frac{\Delta\theta}{\Delta t}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Angular acceleration:

$$\alpha = \frac{d\omega}{dt}, \qquad \alpha_{avg} = \frac{\Delta\omega}{\Delta t}$$

Note that all of the kinematic formulas still work as long as you stay in the same relative frame of reference (rotational or linear). You cannot "mix and match" the two sets of equations.

You may be asking, "If the object is moving at a constant speed, why is there acceleration?" Recall that acceleration is a vector quantity, meaning it has both magnitude and direction. Even though the speed (the magnitude of the velocity) is constant, the direction of the velocity vector is always changing. Since acceleration is the rate of change of velocity, any change in direction means there is acceleration. This acceleration is always directed toward the center of the circle, which is why it is called centripetal (center-seeking) acceleration.

Now that we've established the basic relationships for circular motion, let's apply them to a practical example.

8.2 The Flying Billiard Ball

Let's say you tie a 0.16 kg billiard ball to a long string and begin to swing it around in a circle above your head. The string, perpendicular, is 3 meters long, and the ball returns to where it started every 4 seconds. We will assume the ball moves at a constant speed. If you start your stopwatch as the ball crosses the x-axis, the coordinates of the position (x, y, z) of the ball at any time t given by:

$$p(t) = \left[3\cos\left(\frac{2\pi}{4}t\right), 3\sin\left(\frac{2\pi}{4}t\right), 2\right]$$

(This assumes that the ball would be going counter-clockwise if viewed from above. The spot you are standing on is considered the origin $[0, 0, 0]$.)

Notice that the height is a constant — 2 meters in this case. That isn't very interesting, so we will talk just about the first two components. Figure 8.1 shows a visual of the situation:

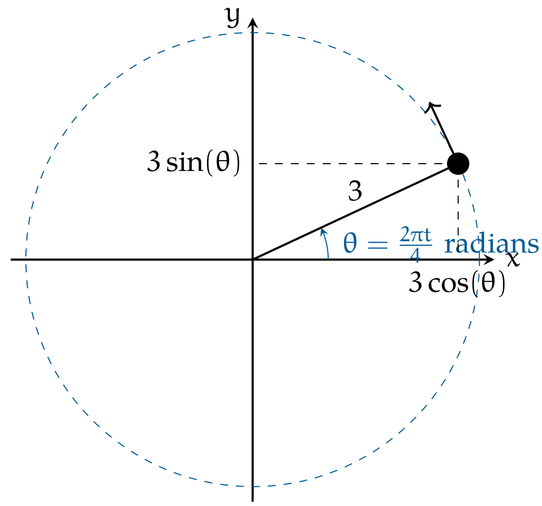


Figure 8.1: A diagram of our ball and string scenario.

In this case, the radius, r, is 3 meters. The period, T, is 4 seconds. In general, we say that circular motion is given by:

$$p(t) = \left[r \cos \frac{2\pi t}{T}, r \cos \frac{2\pi t}{T} \right]$$

A common question is "How fast is it turning right now?" If you divide the 2π radians of a circle by the 4 seconds it takes, you get the answer "About 1.57 radians per second." This is known as *angular velocity* and we typically represent it with the lowercase Omega: ω. (Yes, it looks a lot like a "w".) To be precise, in our example, the angular velocity is $\omega = \frac{\pi}{2}$. Note that in this scenario, the angular velocity and linear *speed* are constant. However, the *velocity* vector is not constant; as we will see in the next section, the direction of the velocity vector is always changing.

Notice that this is different from the question "How fast is it going? (referring to *linear velocity*)" This ball is traveling the circumference of $6\pi \approx 18.85$ meters every 4 seconds. This means the speed of the ball is about 4.71 meters per second.

It is very important to distinguish between angular velocity and linear velocity. Angular velocity is how fast the angle is changing and is referred to by ω, while linear velocity, v, is how fast the object is moving along its path. While linear velocity has a constant speed, it has always changing direction. The angular velocity is constant, but the linear velocity is not.

8.3 Velocity

The velocity of the ball is a vector, and we can find that vector by differentiating each component of the position vector.

For any constants a and b:

Expression $f(x)$	Derivative $f'(x)$
$a \sin bt$	$ab \cos bt$
$a \cos bt$	$-ab \sin bt$

Thus, in our example, the velocity of the ball at any time t is given by:

$$v(t) = \left[-\frac{3(2\pi)}{4} \sin \frac{2\pi t}{4}, \frac{3(2\pi)}{4} \cos \frac{2\pi t}{4}, 0\right]$$

Notice that the velocity vector is perpendicular to the position vector. It has a constant magnitude.

In general, an object traveling in a circle at a constant speed has the velocity vector:

$$v(t) = [-r\omega \sin \omega t, r\omega \cos \omega t]$$

where $t = 0$ is the time that it crosses the x axis. If ω is negative, that means the motion would be clockwise when viewed from above.

The magnitude of the velocity vector is $r\omega$. See Figure 8.2 for a visual of the velocity vector. Note that the z-coordinate is constant, so its derivative is zero, and it is eliminated from our vector for simplification.

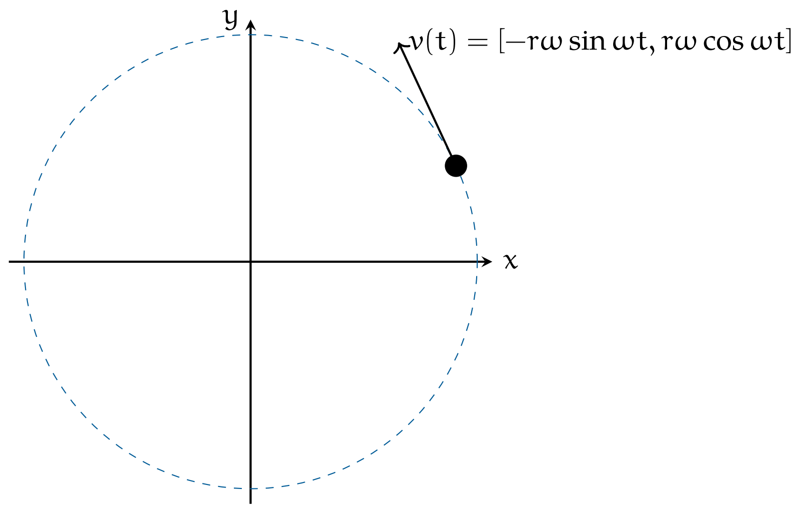


Figure 8.2: Velocity vector of the ball in circular motion.

Exercise 23 A Particle in Motion

[This question was originally presented as two multiple-choice problems on the 2012 AP Physics C exam.] The x and y coordinates of a particle as it moves in a circle are given by:

$$x = 5\cos(3t) y = 5\sin(3t)$$

What is the radius of the particle's circular path? What is the particle's *speed*? Based on your answers, how long does it take the particle to complete the circular path?

Working Space

Answer on Page 122

8.4 Acceleration

We can get the (linear) acceleration by differentiating the components of the velocity vector.

$$a(t) = \left[-r\omega^2 \cos\omega t, -r\omega^2 \sin\omega t\right] = -r\omega^2 [\cos\omega t, \sin\omega t]$$

Notice that the acceleration vector points **toward the center** of the circle it is traveling on. That is, when an object is traveling on a circle at a constant (uniform) speed (notice that coefficients r and ω are constant), its only acceleration is toward the center of the circle. The acceleration does not come from the motion itself, it comes from the constantly changing direction. This is known as *centripetal acceleration*.

The magnitude of the acceleration vector is $a_c = r\omega^2$, or more commonly, $a_c = \frac{v^2}{r}$. This gives us the common calculation for whatever force acts as the centripetal force: $F_c = ma_c = \frac{mv^2}{r}$.[1]

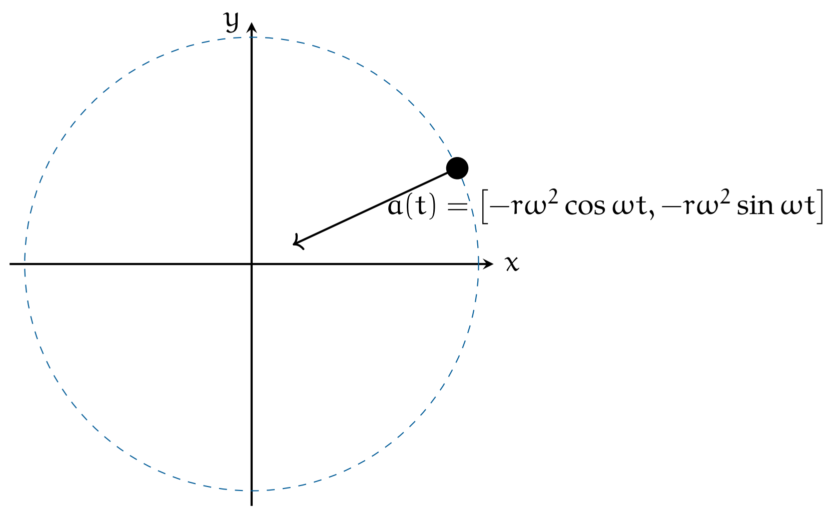
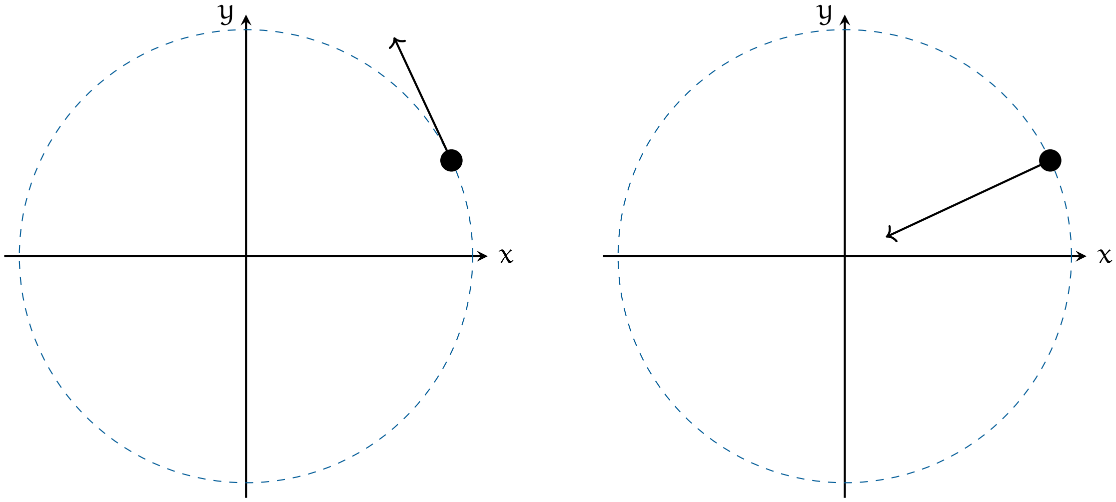


Figure 8.3: Acceleration vector of the particle in circular motion.

Let's compare the velocity and acceleration paths from figures 8.2 and 8.3. Notice that the velocity vector is always tangent to the circle, while the acceleration vector is always pointing toward the center of the circle.

[1]You will see centripetal acceleration noted as a_c or a_r, standing for centripetal or radial, respectively. Note that both refer to the same acceleration.



(a) Velocity vector. (b) Acceleration vector.

Figure 8.4: Uniform circular motion: velocity (tangent) and acceleration (centripetal).

8.5 Centripetal force

What would happen if there were no force pulling the ball toward the center of the circle? It would fly off in a straight line, according to Newton's First Law.

The force you are exerting on the string is what causes it to accelerate toward the center of the circle. We call this the *centripetal force*. It makes sense that the centripetal force and centripetal acceleration are in the same direction, as they are proportional.

Recall that $F = ma$. The magnitude of the acceleration is $r\omega^2 = 3\left(\frac{2\pi}{4}\right)^2 \approx 7.4\text{m/s}^2$. The mass of the ball is 0.16 kg. So, the force pulling against your hand is about 1.18 newtons.

The general rule is that when something is traveling in a circle at a constant speed, the centripetal force needed to keep it traveling in a circle is:

$$F = mr\omega^2$$

If you know the radius r and the speed v of the object, here is the rule:

$$F = \frac{mv^2}{r}$$

We didn't introduce centripetal force as a type of force because it isn't its own force, like friction or gravity. Rather, calling a force "centripetal" tells us what the force is doing. A centripetal force is any force that causes circular motion, a force which acts in the *radial* direction. For a satellite, the centripetal force is gravity. In the opening example with the billiard ball, the centripetal force is the tension in the string.

Example: A child is sitting on a merry-go-round as it spins. What provides the centripetal force?

Solution: In this case, the child is rotating horizontally, so the centripetal force must also be horizontal. Therefore, the centripetal force isn't the child's weight or the normal force. It is the *friction* between the child and the merry-go-round that keeps the child turning. Here is a free body diagram:

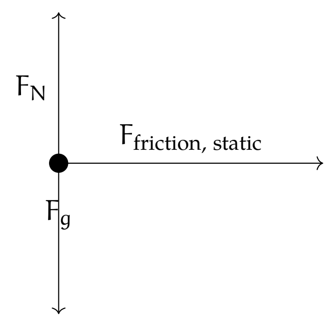


Figure 8.5: The static friction is what causes the centripetal force.

Example: If the child is 1.2 meters from the center of the merry-go-round and it spins at 0.25 rotations/second, what is the coefficient of friction between the child and the merry-go-round?

Solution: Using our FBD and applying Newton's Second Law, we see that:

$$F_N - F_g = 0 \rightarrow F_N = m_{child}g$$

$$F_f = mr\omega^2 = \mu F_N = \mu mg$$

Since the child isn't accelerating vertically, we know that the normal force equals the child's weight. The horizontal force, friction, must equal the mass of the child times the acceleration. Additionally, from the vertical component, we know the friction is equal to μmg, since the normal force is equal to the child's weight. Looking at the second equation, we see that:

$$mr\omega^2 = \mu mg$$

We were given ω in rotations per second, but we need radians per second:

$$\frac{0.25\text{rotation}}{1\text{second}} = \frac{1\text{rotation}}{4\text{seconds}} = \frac{2\pi\text{radians}}{4\text{seconds}} = \frac{\pi}{2}\frac{rad}{s}$$

We can divide m from both sides of $mr\omega^2 = \mu mg$ (notice: we don't need to know the child's mass to determine the coefficient of friction!):

$$r\omega^2 = \mu g \rightarrow \mu = \frac{r\omega^2}{g} = \frac{(1.2\text{m})\left(\frac{\pi}{2\text{s}}\right)^2}{9.8\frac{\text{m}}{\text{s}^2}} = 0.302$$

Therefore, the coefficient of friction between the child and the merry-go-round is 0.302.

8.5.1 Banked Turns

Have you ever been driving on the highway and taken a turn where the road is at an angle? Or maybe you've seen a sign like Figure 8.6 on the road as you take a turn.

Figure 8.6: A banked turn sign.

A banked curve is designed so that a car can safely turn without slipping, even in the rain, if the driver does not exceed the indicated speed. Engineers choose an angle such that the bank provides sufficient force to turn the car without friction. Let's look at a free body diagram for a car taking a banked turn (see figure 8.7).

In the past, we've split the vector for the force of gravity into components that are parallel and perpendicular to the ramp. This time, we will split the normal force into x and y components (see figure 8.8).

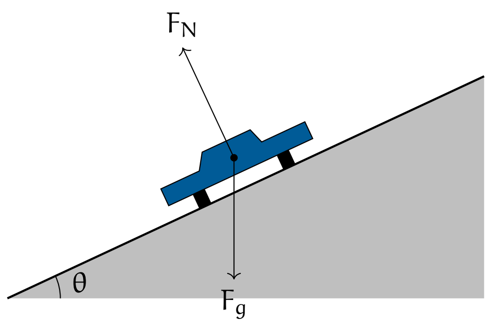


Figure 8.7: If there is no friction due to rain, then the only two forces acting on the car are gravity and the normal force.

Assuming the radius of the turn is 20 m, at what angle should the engineers build the banked turn? (Let's also assume the maximum speed is 25 mph, like the traffic sign above.) First, let's apply Newton's Second Law in the x and y directions:

$$(1)\ F_N \cos(\theta) - F_g = 0$$

$$(2)\ F_N \sin(\theta) = \frac{mv^2}{r}$$

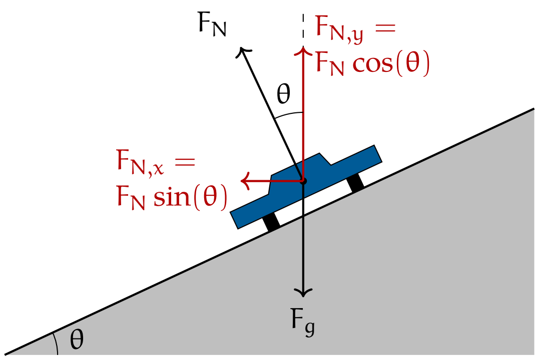


Figure 8.8: Geometrically, the x component of the normal force is given by $F_N \cos(\theta)$ and the y component by $F_N \sin(\theta)$.

We know that $F_g = mg$. From this and equation (1) we see that:

$$F_N = \frac{mg}{\cos(\theta)}$$

Substituting for F_N into equation (2):

$$\left(\frac{mg}{\cos(\theta)}\right)\sin(\theta) = \frac{mv^2}{r}$$

The mass can be divided from both sides, and $\frac{\sin(\theta)}{\cos(\theta)} = \tan(\theta)$:

$$g \tan(\theta) = \frac{v^2}{r}$$

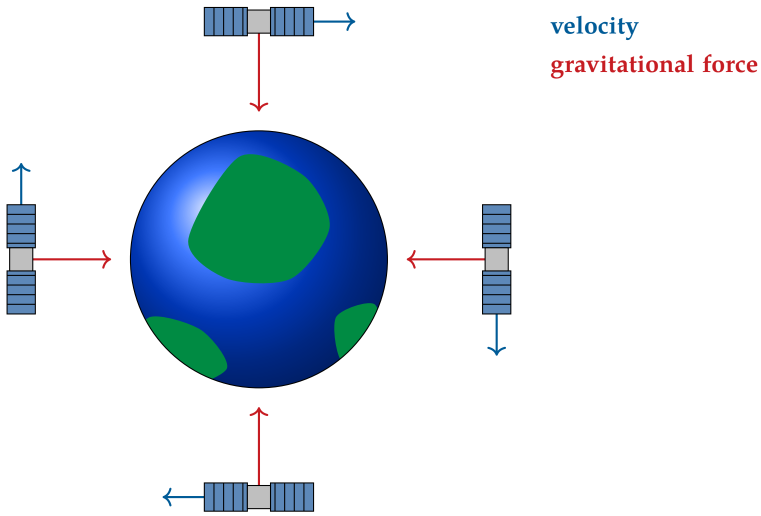
Solving for θ and substituting for the speed ($25\ \text{mph} \approx 11.18\frac{m}{s}$) and radius:

$$\theta = \arctan\left(\frac{v^2}{gr}\right) = \arctan\left(\frac{(11.18\frac{m}{s})^2}{(9.8\frac{m}{s^2})(20m)}\right)$$

$$\theta = \arctan(0.6377) \approx 32.53°$$

8.6 Modeling Circular Motion

What causes circular motion is a constant force perpendicular to the motion. Consider a satellite circling the Earth. The only force acting on the satellite is gravity, yet the satellite does not fall to the Earth. Why? Let's look at the relative direction of motion and gravity for some different positions of the satellite (we'll assume this satellite is moving clockwise from our point of view):



No matter what position the satellite is in, the velocity and gravity vectors are perpendicular.

Exercise 24 Circular Motion

Just as your car rolls onto a circular track with a radius of 200 m, you realize your 0.4 kg cup of coffee is on the slippery dashboard of your car. While driving 120 km/hour, you hold the cup to keep it from sliding.

What is the maximum amount of force you would need to use? (The friction of the dashboard helps you, but the max is when the friction is zero.)

Working Space

Answer on Page 123

Exercise 25 Twirling a Whistle

The lifeguard at a local pool is twirling their whistle horizontally. You wonder if the lifeguard could spin the whistle fast enough to break the string. The string the whistle is attached to can hold a maximum mass of 20 kg before breaking. If the lifeguard's whistle string is 0.35 m long and the average whistle has a mass of 165 grams, what is the maximum tangential speed the lifeguard can spin the whistle? How many rotations per second would the whistle be spinning at? Based on this, do you think the lifeguard is capable of spinning the whistle fast enough to break the string?

Working Space

Answer on Page 123

Exercise 26 The Gravitron

The Gravitron is a carnival ride where riders "stick" to the wall of a spinning cylinder as the floor beneath them drops away. A video explanation is given here: `https://www.youtube.com/watch?v=ifAY5tbYDmQ`. Draw a free body diagram of a rider. If the coefficient of friction between a rider and the wall is 0.32 and the ride is 10 meters across, what angular velocity must the ride reach before the floor drops away?

Working Space

Answer on Page 124

8.7 Flying Pigs and Tension

Take a look at the following two videos that involve flying pigs!

1. Flying Pig: Uniform Circular Motion and Centripetal Force
2. flying pig - circular motion [workthrough demo]

This involves a combination of circular motion and tension. Let's say we have a pig at making angle θ with the vertical and the string of length L, causing the pig to move in a circular path with radius r. We can draw the following two diagrams for our pig, as shown in Figure 8.9.

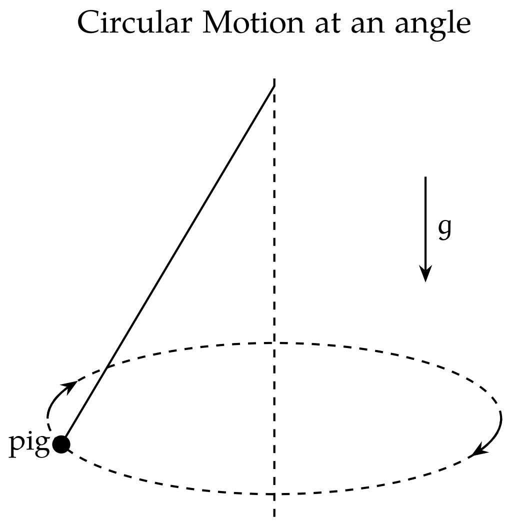


Figure 8.9: The uniform motion of a flying pig attached to a string.

If this is confusing, think of a tetherball from your local playground. It follows the same mechanics!

At any point, we can draw a free body diagram for the pig. Figure 8.10 shows the forces acting on the pig at all points.

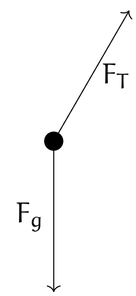


Figure 8.10: The FBD of the flying pig at any point.

To analyze the net forces, the tension force needs to be split into its x and y components. The y component of the tension force must balance out the weight of the pig (as it is not moving vertically), while the x component of the tension force provides the centripetal force to keep the pig moving in a circle.

$$\sum F_y = 0 \rightarrow F_{T_y} - F_g = 0 \rightarrow F_{T_y} = F_g = F_T \cos\theta = mg$$

The centripetal force is found by:

$$F_{T_x} = F_c = F_T \sin\theta = ma_r = \frac{mv^2}{r}$$

Dividing the second equation by the first equation gives:

$$\frac{F_T \sin(\theta)}{F_T \cos(\theta)} = \frac{\frac{mv^2}{r}}{mg} \rightarrow \tan(\theta) = \frac{v^2}{rg}$$

Solving for θ gives us $\theta = \arctan\left(\frac{v^2}{rg}\right)$. We can also solve for v:

$$v = \sqrt{rg\tan\theta} = \sqrt{Lg\sin\theta\tan\theta}$$

Since r can be solved for θ using $r = L\sin\theta$.

Explicitly, we can express our tension force as $F_T = \frac{mg}{\cos(\theta)}$ (coming from our net force in the y-direction).

We can also explicitly solve for ω:

$$\begin{aligned}
\tan(\theta) &= \frac{v^2}{rg} \\
&= \frac{(r\omega)^2}{rg} = \frac{r\omega^2}{g} \\
\omega^2 &= \frac{g\tan(\theta)}{r} \\
\omega &= \sqrt{\frac{g\tan(\theta)}{r}}
\end{aligned}$$

What does all of this tell us?

- The vertical component of tension balances the weight of the pig, so there is no vertical motion.
- The horizontal component of tension provides the centripetal force that keeps the pig moving in a circle.
- The faster the pig moves, the larger the angle θ and the greater the tension in the string.
- The tension is always greater than the pig's weight: $F_T = \frac{mg}{\cos\theta}$.
- The angular speed required for a given angle is $\omega = \sqrt{\frac{g\tan\theta}{r}}$. This increases with larger angles, and decreases with a shorter radius.

Exercise 27 Playground Tetherball

A tetherball of mass .5kg is attached to a string of length 2.5m. When the ball is swung around in a horizontal circle, the string makes an angle of 38° with the vertical. If the ball is in uniform circular motion, what is the speed of the ball?

Working Space

Answer on Page 125

8.8 Non-uniform circular motion

We have talked enough about uniform circular motion, where the speed is constant. However, in many real-world scenarios, the speed is not constant. This is known as *non-uniform circular motion*. In this case, there are two components of acceleration: the centripetal acceleration $a_\perp$ (toward the center of the circle) and the tangential acceleration $a_\parallel$ (along the direction of motion). The tangential acceleration is responsible for changes in speed, while the centripetal acceleration is responsible for changes in direction. We will expand on this in the oscillations chapter, but this acceleration is caused by a *restoring force*, a force that tries to pull the object back toward equilibrium whenever it's displaced. We will introduce this concept here, but go into more detail in the oscillations chapter.

Let's look at a few examples of these concepts in action.

8.8.1 Roller Coaster Loop

Now we can talk about a roller coaster. A roller coaster loop is a great example of circular motion. As the roller coaster cart goes through the loop, it experiences both centripetal and tangential acceleration.

We will analyze the cart on a roller coaster at positions at the top, bottom, and sides of the loop, as shown in Figure 8.11

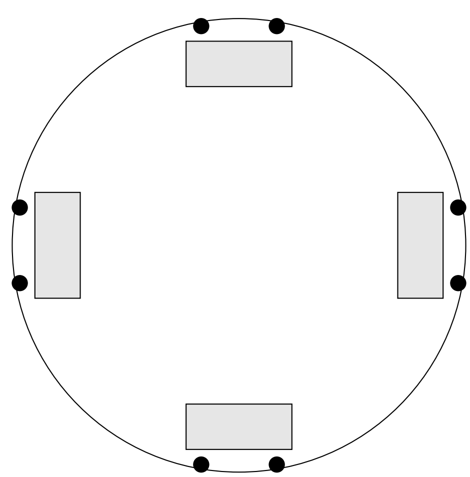


Figure 8.11: Carts at four positions around the loop with wheels on the track.

At the bottom

At the very bottom of the loop, the forces acting on the cart are gravity and the normal force from the track. Let's imagine the cart goes around the circular track infinitely without "exiting" the loop. There must, then, be a force keeping the cart moving in a circle. This force is the centripetal force, which points toward the center of the loop. The only two forces acting on the cart are gravity (downward) and the normal force from the track (upward). We know that a centripetal force must satisfy the formula $F_{net} = F_N - F_g = \frac{mv^2}{r}$. Therefore, the normal force must be greater than the weight of the cart at the bottom of the loop: $F_N = \frac{mv^2}{r} + mg$. This is represented in Figure 8.12.

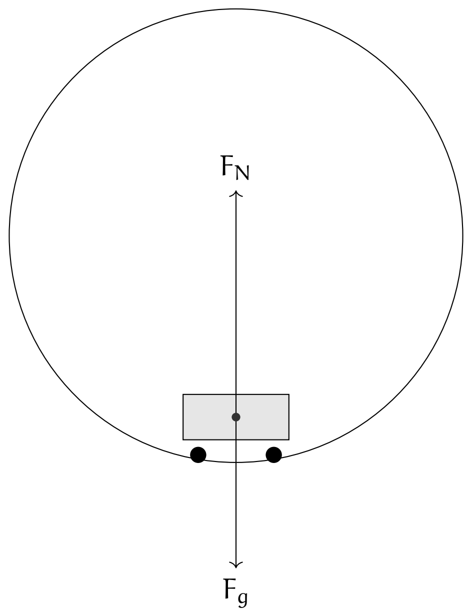


Figure 8.12: The FBD of the roller coaster at the bottom.

At the top

At the very top of the loop, the forces acting on the cart are gravity and the normal force from the track. Both forces point *downward* this time, toward the center of the loop. We can say that $F_{net} = F_N + F_g = \frac{mv^2}{r}$. Solving for the normal force gives: $F_N = \frac{mv^2}{r} - mg$.

Notice that the normal force is less than the weight of the cart at the top of the loop. This means that the centripetal acceleration would have to be greater than or equal mg. This is represented in Figure 8.13.

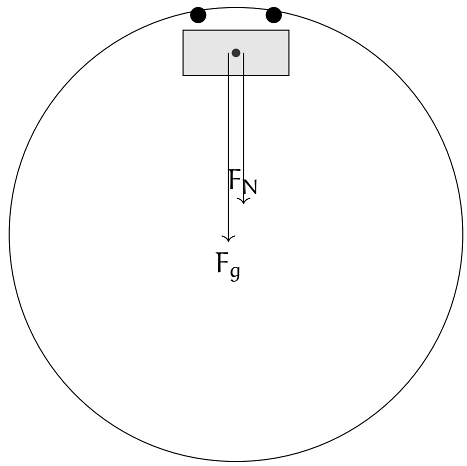


Figure 8.13: The FBD of the roller coaster at the top.

Note that for both the top and bottom of the loop, the minimum speed required to maintain contact with the track is given by $v = \sqrt{rg}$. If the cart goes any slower such that $v < \sqrt{rg}$, it will lose contact with the track, as there will not be enough normal force to provide the centripetal force needed to keep the cart moving in a circle. Think about where this observation comes from!

At the sides

Now at the sides of the loop (say 0 and π if we view the roller coaster as a unit circle), the net force is directed at an angle. The normal force from the track acts horizontally, pointing toward the center of the circle, while the gravitational force acts vertically downward. Since these two forces are perpendicular to each other, the net force is the vector sum of the normal and gravitational forces.

At the right side ($\theta = 0$), the normal force points leftward (toward the center), and gravity points downward, so the net force points diagonally down and to the left, toward the lower-left quadrant of the circle.

At the left side ($\theta = \pi$), the normal force points rightward (toward the center), and gravity still points downward, so the net force points diagonally down and to the right, toward the lower-left quadrant of the circle.

While the normal force provides the centripetal component of the net force, gravity introduces a tangential component, causing the cart to accelerate (speed up) along the track. Since the centripetal force is not directly towards the center of the circle, the cart must be speeding up. This is modeled in Figure 8.14.

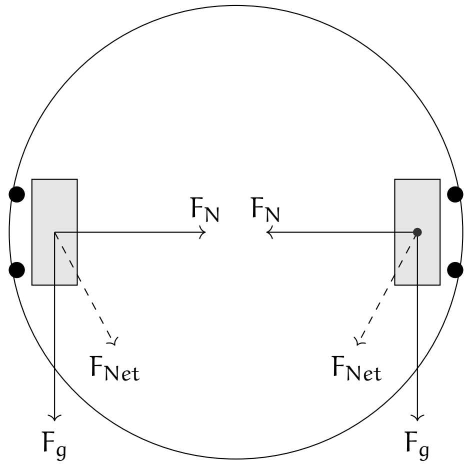


Figure 8.14: The FBD of the roller coaster at the sides.

Exercise 28 Roller Coaster Loop

This question is taken from the AP Physics C Review Book, 19th edition.

A roller-coaster car enters the circular loop portion of the ride. At the very top of the circle (where people are upside down), the car has a speed of 25 m/s, and the acceleration points straight down. If the diameter of the loop is 50 m and the total mass of the car plus passengers is $1,200$ kg, find the magnitude of the normal force exerted by the track on the car at this point. Also find the normal force exerted by the track on the car when it is at the very bottom of the loop, given it is still traveling at 25 m/s. You may use $g = 10\ \text{m/s}^2$ for simpler calculations.

Working Space

Answer on Page 126

Our next chapter will explore oscillations, restoring forces, simple harmonic motion, and more concepts of harmonic motion.

Chapter 9

Oscillations

We just finished up a chapter about circular motion, where objects move around in a circle at either constant (uniform/harmonic) or changing (non-uniform) speeds.

In this chapter, we will study oscillatory motion, which is a type of motion where an object moves back and forth around an equilibrium point. Oscillations are common in many physical systems, such as springs, pendulums, and even electrical circuits. Oscillations are important not only because they appear everywhere in physical systems, but also because they provide a connection between simple motion and more complex behaviors. Small oscillations often behave in a very predictable way and can be described using *simple harmonic motion* (SHM), which is governed by restoring forces proportional to displacement.

We will also give an introduction to *differential equations* used to model oscillatory systems, often found in engineering and physics.

9.1 Sine and Cosine Graphs [Review]

Let's recall the two pillars of trigonometry and, consequently, simple harmonic motion: Sine and Cosine. Recall that these two functions are periodic, meaning they repeat their values in regular intervals or periods. The sine and cosine functions have a period of 2π radians or $360°$, repeating their values every 2π radians.

As discussed in the circular motion chapter, the sine function represents the y-coordinate of a point on the unit circle as it moves around the circle, while the cosine function represents the x-coordinate. This is the fundamental behavior of the unit circle, where a point moves around the circle at a constant angular speed ω. The position of the point at any time t can be described using the following parametric equations:

$$x(t) = \cos(\omega t), \qquad y(t) = \sin(\omega t)$$

The general forms of the sine and cosine functions can be expressed as:

$$x(t) = A\cos(\omega t + \phi), \qquad y(t) = A\sin(\omega t + \phi)$$

where:

- A is the amplitude, representing the maximum displacement from the equilibrium position.
- ω is the angular speed, which determines how quickly the oscillations occur.
- ϕ is the phase shift, which determines the initial position of the oscillation at $t = 0$.

Recall that the derivatives of these functions are also periodic, a key property that is useful for analyzing oscillatory motion:

$$\frac{d}{dt}\cos(\omega t) = -\omega \sin(\omega t)$$
$$\frac{d}{dt}\sin(\omega t) = \omega \cos(\omega t)$$
$$\frac{d^2}{dt^2}\cos(\omega t) = -\omega^2 \cos(\omega t)$$

In the motion we will talk about,

- $x(t)$ will be our position function
- $v(t)$ describes our velocity function, which can be the first derivative of our position
- $a(t)$ is our acceleration function, derived from $v(t)$

We notice that the second derivative produces a negative multiple of the original function. This can be simplified in terms of acceleration and position functions: $a(t) = -\omega^2 x(t)$.

9.2 Simple Harmonic Motion

Simple Harmonic Motion will reference many definitions we have seen before and will continue to see in this chapter. Here are some key terms to know:

Cycle A single complete execution of a periodically repeated motion. Another way to think about it is a particle, block, or object making a *round-trip*.

Period The time it takes to complete one cycle of motion. An identity that comes up often is $T = \frac{2\pi}{\omega}$.

Frequency The number of cycles completed per unit time. It is the reciprocal of the period, represented as **Hz**.[1]

[1] One **Hz** is equal to one cycle per second. For example, a frequency of 5 Hz means that 5 cycles are completed in one second. On a monitor, a refresh rate of 60 Hz means the screen updates 60 times per second. Higher refresh rates can lead to smoother motion perception.

Equilibrium The position at which the "default" position of an oscillating object is located. In terms of mechanics, this is the position where all the forces acting on the object are balanced or zero.

Restoring Force The force that acts to bring an oscillating object back to its equilibrium position. It is typically proportional to the displacement from equilibrium and acts in the opposite direction.

Angular Speed The rate at which an object moves around a circle, measured in radians per second.

So far, we have discussed oscillatory motion in general terms. Now, let's focus on a specific type of oscillatory motion known as Simple Harmonic Motion (SHM). SHM is characterized by the following properties:

- The motion occurs around an equilibrium position.
- The *restoring force* is directly proportional to the displacement from the equilibrium position and acts in the opposite direction.
- The motion is sinusoidal in nature, meaning it can be described using sine or cosine functions.
- The period and frequency of the motion are constant, regardless of the amplitude.
- The acceleration always points toward equilibrium and increases in size as the displacement grows.

So the motion continuously repeats itself in a regular pattern. It speeds up near equilibrium, and slows down to a stop at maximum displacement. Examples of these systems are

- A mass attached to a spring, either vertically or horizontally
- A simple pendulum
- Vibrations of a tuning fork or guitar string

We saw in our pendulums chapter how the system has both centripetal acceleration and acceleration due to gravity, causing oscillations back and forth with changing speed. Like all mechanical systems, an SHM system has both kinetic and potential energy. For general simple harmonic motion systems, we can describe its energy as follows:

- The total mechanical energy E of the system is the sum of its kinetic energy K and potential energy U:
$$E = K + U$$
- The kinetic energy K, system dependent, is highest when the object passes through

the equilibrium position.

- The potential energy U depends on the specific system, such as spring potential energy or gravitational potential energy, but is highest at the maximum displacement from equilibrium.

9.3 Springs

Let's talk about springs! Springs are mechanical devices, usually made out of some rigid metal material, that store and release energy through deformation. When a spring is stretched or compressed from its equilibrium position, it exerts a restoring force that tries to bring it back to its original shape. The most common type of spring is a coil spring, which consists of a linear wire would into a helical shape.

9.3.1 Hooke's Law

In 1676, physicist Robert Hooke formulated a principle that describes the behavior of springs, now known as Hooke's Law. Hooke's Law states that the force exerted by an ideal spring is directly proportional to the displacement from its equilibrium position.

Mathematically, Hooke's Law can be expressed as:

$$F_s = -kx$$

where:

- F_s is the restoring force exerted by the spring (in newtons, N)
- k is the spring constant (in newtons per meter, N/m), which measures the stiffness of the spring. It is spring-specific
- x is the displacement from the equilibrium position (in meters, m)
- The negative sign represents the fact that the force attempts to bring the spring back to its equilibrium position.

Since we know that force is related to mass and acceleration through Newton's Second Law, we can combine Hooke's Law with Newton's Second Law to analyze the motion of a mass attached to a spring. See Figure 9.1 for a diagram of a mass-spring system.

$$F_s = ma = -kx \tag{9.1}$$

$$ma = -kx \tag{9.2}$$

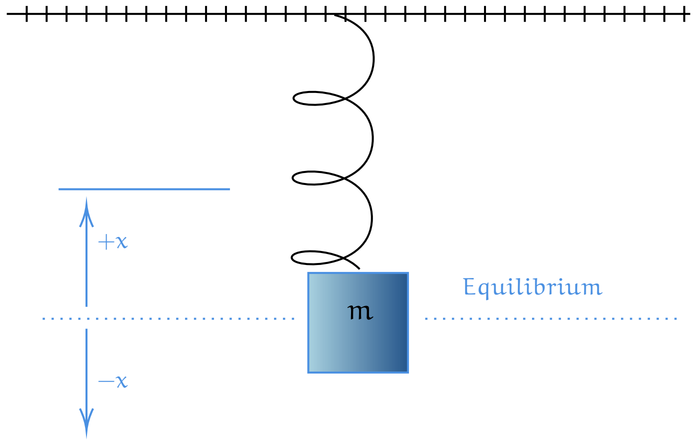


Figure 9.1: A diagram showing a block of mass m displaced $+x$ and $-x$ from equilibrium.

But since above we found a relation between acceleration and position, we can write:

$$m\frac{d^2x}{dt^2} = -kx \tag{9.3}$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \tag{9.4}$$

We can see that this equation describes simple harmonic motion, where the acceleration is proportional to the negative of the displacement. In this form, we can identify the angular speed ω of the oscillation as:

$$\omega = \sqrt{\frac{k}{m}}$$

as the term in front of the x in equation (9.4) is ω^2. Thus, we can further define the period as

$$T = 2\pi\sqrt{\frac{m}{k}}$$

and the frequency as

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

In a later section of this chapter, we will solve this differential equation to find the position function $x(t)$ of general mass-spring systems.

Exercise 29 Mass-Spring Frequency Change

A block is attached to a spring and set into oscillatory motion, and its frequency is measured. If this block were removed and replace with a block of $1/2$ the mass, how would the frequency of oscillations compare to the original frequency?

1. The frequency would be halved.
2. The frequency would remain the same.
3. The frequency would increase by a factor of $\sqrt{2}$.
4. The frequency would decrease by a factor of $\sqrt{2}$.

Working Space

Answer on Page 127

9.3.2 Spring Potential Energy

Let's recall that Work from a physics standpoint is defined as the transfer of energy through a certain distance via a force. When we alter the length of a spring through either compression or extension, we are performing work on the spring. This work is stored as potential energy within the spring, which can be released when the spring returns to its equilibrium position.

Let's find an equation for this change in potential energy, which gets stored up in the spring. Recall the two mathematical definitions for work:

- Work as the product of force and distance: $W = F \cdot d$
- Work as the integral of force over a distance: $W = \oint F\,dx$

We need to use the integral definition, since, by our definition of Hooke's Law, the force exerted by a spring varies with displacement, so every tiny bit of distance, or dx is different. For any spring starting at equilibrium (0), and moving to some final position x_f, we can write:

$$\begin{aligned} W &= \int_{x_i}^{x_f} F_x \, dx \\ W_s &= \int_{x_i}^{x_f} F_s \, dx \\ &= \int_{0}^{x_f} -kx \, dx \\ &= \left. \frac{-kx^2}{2} \right|_{0}^{x_f} \\ &= \frac{-k(x_f)^2}{2} \end{aligned}$$

This work done on the spring is stored as potential energy, so we can say that the change in potential energy of the spring is equal to the negative of the work done by the spring force:

$$\Delta U_s = -W_s = \frac{1}{2} k x_f^2 \tag{9.5}$$

We can see this graphically in Figure 9.2, where the area under the force vs. displacement graph represents the work done on the spring, which is equal to the change in potential energy.

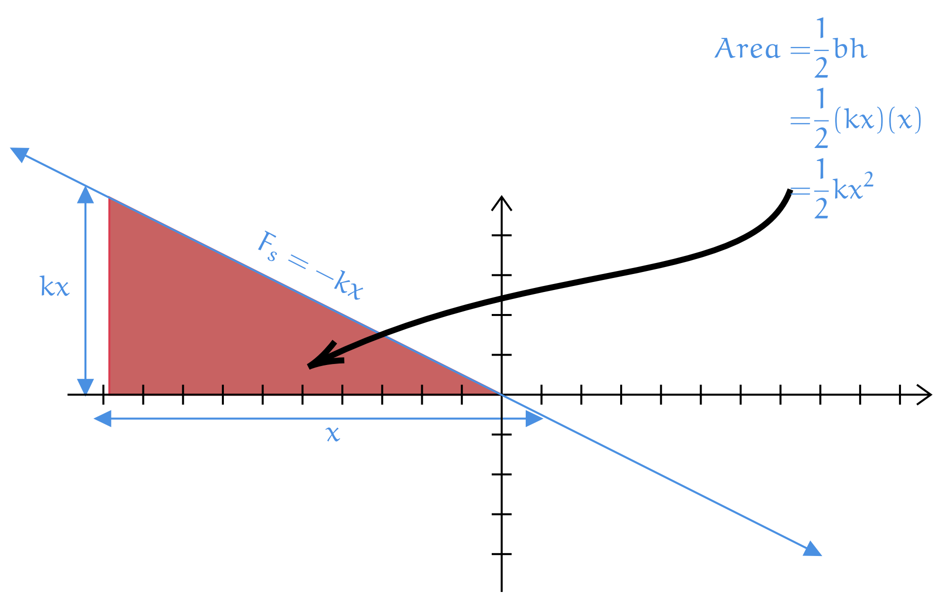


Figure 9.2: A graph of force versus displacement for a spring, equal to the work done and change in potential energy.

Notice that Equation (9.5) is similar in structure to the kinetic energy equation,$K = \frac{1}{2} m v^2$. This is not a coincidence, as both equations describe forms of energy in mechanical systems, and are derived from linear systems. When the effort (force) needed increases in direct proportion to what you're trying to change, the energy required ends up depending on the square of that change. For a spring

- the farther you stretch or compress it, the more force it exerts to return to equilibrium (Hooke's Law)
- the more energy is stored in the spring (spring potential energy)

For kinetic energy

- the faster you try to move an object, the more force is needed to accelerate it (Newton's Second Law)
- the more energy the object has due to its motion (kinetic energy)

Exercise 30 Finding the equation for a mass-spring system

This question is from an AP-style physics review. A block of mass 4 kg on a frictionless, horizontal table is attached to a spring with spring constant $k = 400\,\mathrm{N/m}$ and undergoes simple harmonic oscillations about its equilibrium position and its amplitude $A = 6$ cm. If the block is at $x = 6$ cm at time $t = 0$ and has a velocity of 0 cm/s at that time, which of the following equations (with x in cm and t in seconds) describes the block's position as a function of time?

(A) $x(t) = 6\cos(10t)$

(B) $x(t) = 6\sin(10t)$

(C) $x(t) = 6\sin(10t + \frac{1}{2}\pi)$

(D) $x(t) = 6\sin(10\pi t + \frac{1}{2}\pi)$

(E) $x(t) = 6\sin(10\pi t - \frac{1}{2}\pi)$

Working Space

Answer on Page 127

Exercise 31 Energy Transfer in a Mass-Spring System

A pinball machine uses a spring to launch a ball of mass 0.2 kg. The spring has a spring constant of 500 N/m and is compressed by 0.24 m before release. Assuming energy is conserved and no frictional losses occur, what is the speed of the ball as it leaves the spring?

Working Space

Answer on Page 128

9.4 Mass-Spring Systems and Linear Differential Equations

We have already seen that the motion of a mass attached to a spring can be model e d using different ial equations. Here, we will examine this connection more clo sel y.

9.4.1 Undamped Simple Harmonic Motion

Starting from Newton's Second Law and Hooke's Law, we derived the differential equation for simple harmonic motion:

$$m\frac{d^2x}{dt^2} + kx = 0. \tag{9.6}$$

The term$m\frac{d^2x}{dt^2}$comes directly from *Newton's Second Law*, representing the net force required to accelerate the mass. The term kx arises from *Hooke's Law*, which states that the spring exerts a restoring force proportional to displacement and directed toward equilibrium. Equation (9.6) is an example of a **second-order linear differential equation with constant coefficients without damping** . It describes an ideal mass-spring system with no external influences: one that oscillates forever, undamped and unforced. The $ma(t)$ term is *Newton's Second Law* part of the equation, while the $kx(t)$ term comes from *Hooke's Law*. This is a second-order linear differential equation with constant coefficients. What if the spring system is be ing driven by a third force, like friction or damping? We can add an additional force term$F(t)$to the equation:$F_f(t) = -cv = -cx'$ for forcing function.

9.4.2 Introducing Damping

Real systems are often influenced by additional forces. One common example is a **damping force**, such as friction or air resistance, which opposes the motion of the mass. A typical damping force is proportional to the velocity and can be written as

$$F_d(t) = -c\,\frac{dx}{dt},$$

wherecis the damping coefficient. Including this force in Newton's Second Law modifies our differential equation. Instead of the undamped equation (9.6), we obtain:

$$ma(t) = -c\,v(t) - k\,x(t) \tag{9.7}$$

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0 \tag{9.8}$$

This is still a second-order linear differential equation with constant coeffici e nts, but its solutions behave very differently. Depending on the value of c, th e system may oscillate with decreasing amplitude, fail to oscillate at all, or re t urn to equilibrium as quickly as possible. Finding the roots of the characteristic equation associated with Equation (9.8) allows us to classify the system's behavior into three categories. Recall that the quadratic formula gives us the roots using our coefficients:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

Characteristic Roots	Homogeneous Solution	Damping Type
$r_1, r_2 \in \mathbb{R}$, both negative	$y_h = C_1e^{r_1t} + C_2e^{r_2t}$	Overdamped
r real, repeated	$y_h = (C_1 + C_2t)e^{rt}$	Critically damped
$r = \alpha \pm i\beta$, $\alpha < 0$	$y_h = e^{\alpha t}(C_1\cos(\beta t) + C_2\sin(\beta t))$	Underdamped
$r = \pm i\beta$	$y_h = C_1\cos(\beta t) + C_2\sin(\beta t)$	Simple Harmonic Motion

Table 9.1: Classification of solutions to the damped harmonic oscillator based on characteristic roots.

Without going into too much linear algebra, the function of a mass spring is giv e n by$y(t) = y_h(t) + y_p(t)$, where y_h is the homogeneous solution (the p art we solved above by setting the differential equation to 0), and y_p is the p articular solution, which depends on any external forcing functions. In this section, we focused on the homogeneous solution. If the equation (9.8) had a forcing function$F(t)$on the right side, we would need to find a particular solutiony_p to account for that.

Exercise 32 Linear Systems 1

Working Space

A 2 kg mass is attached to a spring with $k = 200\,\text{N/m}$ with no damping. Which differential equation describes the motion?

1. $x'' + 100x = 0$
2. $x'' + 10x = 0$
3. $x'' + 200x = 0$
4. $2x'' + 100x = 0$

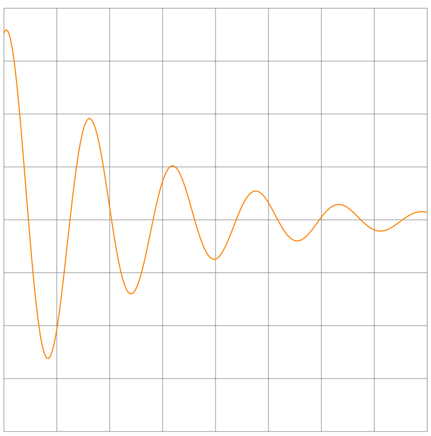
Answer on Page 128

Exercise 33 Linear Systems 2

Work

Answer the following questions about the graph below, which shows the position of a mass on a spring as a function of time. The equation of motion for this system is given by:

$$x(t) = 0.15e^{-0.40t}\cos(4t + 0.30)$$



- Is this system undergoing simple harmonic motion, or is it damped? If it is damped, is it underdamped, overdamped, or critically damped?
- What would need to be changed in the equation of motion to make this system undergo simple harmonic motion instead?

Answer on Page 129

Exercise 34 Linear Systems 3

Working Space

Given that a unit (1 kg) mass obeys:

$x'' + 16x = 0, \quad x(0) = 0.20, \quad v(0) = x'(0) = 0$

1. Solve for $x(t)$
2. Find the acceleration, $a(t)$, and the velocity, $v(t)$, as functions of time.

Answer on Page 129

9.5 Pendulums

You may be surprised to know that pendulums act the exact same way as roller coasters from the circular chapter, in terms of forces. But, similar to springs, they oscillate back and forth unless acted upon by a damping force. A pendulum has an equilibrium position$\theta = 0$such that it is parallel to its gravitational component. A pendulum also has a maximum angle$\theta_{max}$from its equilibrium position. It will swing through a set arc in a repetitive motion, which is a type of motion we call simple harmonic motion, similar to springs (which we will also cover in the oscillations chapter). The tension force is always directed towards the pivot. This is diagramed below in Figure 9.3

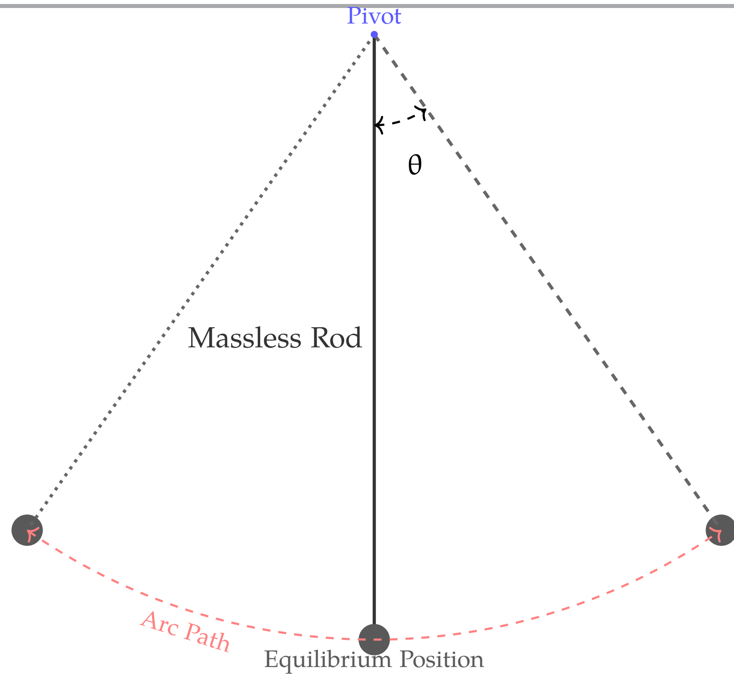


Figure 9.3: The anatomy of a pendulum.

When displaced by an angle$\theta$from the vertical equilibrium position ($\theta = 0$), gravity provides a *restoring force*that acts along the arc of motion. This restoring force is given by$F = -mg\sin\theta$, where the negative sign indicates that the force always points back toward the equilibrium position. The tension in the string keeps the bob moving along its curved path, while gravity continually tries to return it to rest. The net force shifts such that the acceleration arrow is *not*always pointing to the center. The net force is a combination of the tension (centripetal) force and the gravitational (tangential), so rather it points at different points along the arc of the circle, as it is always changing due to the height of the pendulum. This is demonstrated in Figure 9.4. Note the following differences in energy and forces at the equilibrium position versus the maximum angle positions, summarized in Table 9.2. Note here that the magnitude of the restoring

At $\theta = 0$	**At** $\theta = \theta_{max}$
$F_{restoring} = 0$ a_t (tangential acceleration) $= 0$ $PE = 0$ $KE = max$ $V = max$	$F_{restoring} = max$ $a_t = max$ $PE = mgh$ $KE = 0$ $v = 0$

Table 9.2: Comparison of pendulum quantities at equilibrium and maximum displacement

force$-mg\sin(\theta)$, which is not *directly* proportional to the angular displacementθ. However, for small angles (typically less than about 1 radian or 180°), we can use the small-angle approximation$\sin(\theta) \approx \theta$(in r adians), and state the restoring force as$F \approx -mg\theta$. This allows us to treat the pendulum's motion as simple harmonic motion for small oscillations, where the restoring force is approximately proportional to the angular displacement.

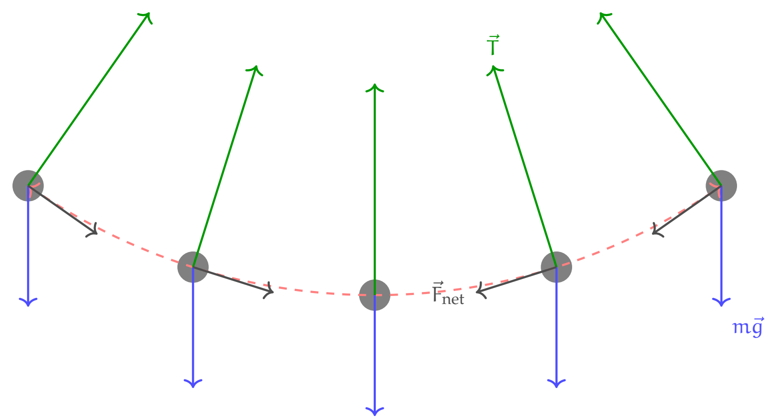


Figure 9.4: The net force at different locations of the arc.

Exercise 35 Pendulum Period Dependence

A simple pendulum consists of a mass m attached to a massless rod L, swinging under the influence of gravity. Derive the equation for the period of a simple pendulum. Which of the following changes would affect the period T of the pendulum's oscillation?

1. Increasing the mass m of the pendulum bob.
2. Increasing the length L of the string.
3. Increasing the amplitude (maximum angle) of the swing.
4. Decreasing the acceleration due to gravity g.

Working Space

Answer on Page 129

This exercise showed you that the period of a simple pendulum depends on the length of the pendulum and the acceleration due to gravity, but not on the mass of the bob or the amplitude of the swing (for small angles). One derivation from the exercise above was the Equation **??**, which can be rearranged as $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$. Solving this differential equation[2]will yield the angular position function$\theta(t) = \theta_{max}\sin(\omega t + \phi_0)$or equivalently,$\theta(t) = \theta_{max}\sin(\sqrt{\frac{g}{L}}t + \phi_0)$of the pendulum over time, which describes its oscillatory motion. Note that this differential equation does not the depend on mass of the bob nor the amplitude of the swingθ_{max}, a core factor of pendulum harmonic motion.

Exercise 36 Pendulum on the Moon

A simple pendulum has a period of 2 seconds when measured on Earth. If the same pendulum were taken to the Moon, where the acceleration due to gravity is approximately $\frac{1}{6}$, what would be its new period?

a Approximately 0.8 seconds

b Approximately 2 seconds

c Approximately 4.9 seconds

d Approximately 6.8 seconds

Working Space

Answer on Page 130

9.6 Alternating Current

We have talked about Alternating Current (AC) in previous chapters, but now we will explore its connection to oscillations. AC is an electric current that periodically reverses direction, in contrast to Direct Current (DC), which flows in a single direction and provides constant voltage. The voltage and current in an AC circuit vary sinusoidally with time, making them ideal for modeling using oscillatory functions.

In Direct Current, $V = V_0$, which means the original voltage is equal to the voltage everywhere else.

In an AC circuit, the voltage at any time t behaves exactly like a mass on a spring, following

[2]We will go more in depth about this in a future chapter.

a sinusoidal oscillation:

$$V(t) = V_{peak}\sin(\omega t + \phi) \tag{9.16}$$

where

- V_{peak} is the maximum voltage reached in either direction
- ω is the angular frequency, related to standard frequency by $\omega = \frac{2\pi}{T} = 2\pi f$
- ϕ which is the phase constant, or the starting position of the wave

Note that, because of the trigonometric nature of sine, the wave fluctuates between V_{peak} and $-V_{peak}$.

For sinusoidal voltages, the root mean squared (RMS) is derived as

$$\begin{aligned}
V_{\text{rms}} &= \sqrt{\frac{1}{T}\int_0^T [V_{\text{peak}}\sin(\omega t + \phi)]^2 dt} \\
&= V_{\text{peak}}\sqrt{\frac{1}{2T}\int_0^T [1 - \cos(2\omega t + 2\phi)]dt} \\
&= V_{\text{peak}}\sqrt{\frac{1}{2T}\int_0^T dt} \\
&= \frac{V_{\text{peak}}}{\sqrt{2}}
\end{aligned}$$

9.6.1 Impedance

In DC, we use Resistance (R). In AC oscillations, we use *Impedance*. Impedance is the "total opposition" to the oscillation and is a combination of standard resistance and Reactance (X).

Capacitors oppose changes in voltage, which we call Capacitive Reactance:

$$X_C = \frac{1}{\omega C}$$

Inductive Reactance comes from inductors which oppose changes in current (acting like "electrical inertia").

$$X_L = \omega L$$

Similarly, in AC circuits, the voltage and current waves do not always peak at the same time.

- In a purely inductive circuit, the current falls behind the voltage by 90° $(\pi/2)$.
- In a purely capacitive circuit, the current leads the voltage by 90° $(\pi/2)$.

9.6.2 Comparing Mass-Springs and AC

Mechanical System (Mass-Spring)	Electrical System (AC Circuit)
Displacement (x)	Charge (q)
Velocity (v)	Current (i)
Mass (m)	Inductance (L)
Spring Constant (k)	Reciprocal Capacitance $(1/C)$
Damping/Friction (b)	Resistance (R)

9.7 Summary

In this chapter, we have discussed oscillations via Simple Harmonic Motion, mass-springs, spring energy, pendulums, and AC currents.

- **Simple Harmonic Motion (SHM)**: A specific type of periodic motion where a restoring force (like a spring) constantly pulls an object back toward an equilibrium point. Key Parameters include:

    Oscillation above and below (or to and from) an equilibrium point.

    A restoring force, proportional to displacement.

    An angular velocity that is always changing, ω

    Time for one full cycle $(T = 2\pi/\omega)$.

- **Hooke's Law**: For an ideal spring, $F_s = -kx$. This linear relationship leads to an angular frequency of $\omega = \sqrt{k/m}$.

- **Differential Equations**: Oscillations can be represented in the form $m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$, where m is the mass, c is the damping constant, and k is the spring constant.

    Underdamped: Oscillates with decaying amplitude.

    Critically Damped: Returns to equilibrium as quickly as possible without oscillating.

    Overdamped: Returns to equilibrium slowly without oscillating.

- **Energy Dynamics**: In a closed system, energy oscillates between Kinetic (K) and Potential (U). Total mechanical energy remains constant $(E = \frac{1}{2}kA^2)$, with K peaking at equilibrium and U peaking at maximum displacement.

- **Acceleration**: Acceleration is affected by both *centripetal acceleration* and *tangential acceleration*.

This chapter leads us into our next chapter on orbits, the oscillations of planetary motion.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 17)

x-intercept: $(-5/2, 0)$; y-intercept: $(0, 5/4)$; horizontal asymptote: $y = 2$; vertical asymptote: $x = -4$

Answer to Exercise 3 (on page 20)

The correct answer is E. To explain, we examine the behavior of each function as $x \to \infty$.
A. $\lim_{x\to\infty} \frac{\sin 5x}{x} = \pm\infty \neq 5$
B. $\lim_{x\to\infty} 5x = \infty \neq 5$
C. $\lim_{x\to\infty} \frac{1}{x-5} = 0 \neq 5$ (this function does have a *vertical* asymptote at $x = 5$).
D. $\lim_{x\to\infty} \frac{5x}{1-x} = \frac{5}{-1} = -5 \neq 5$ (this function has a horizontal asymptote at $x = -5$).
E. $\lim_{x\to\infty} \frac{20x^2-x}{1+4x^2} = \frac{20}{4} = 5$.

Answer to Exercise 3 (on page 20)

Factored form: $\frac{x^2(x+2)}{x(x+1)}$; hole: $(0, 0)$; vertical asymptote: $x = -1$; oblique asymptote: $y = x + 1$

Answer to Exercise 4 (on page 25)

1. $x^2 + 4x + 12\frac{30x-27}{x^2-3x+2}$
2. $2 + \frac{6x^2-4x+13}{x^3-3x^2+2x-4}$
3. $3x - 2 + \frac{8x^2-6x+1}{x^3-3x}$

Answer to Exercise 5 (on page 29)

1. $\frac{10}{7(x+6)} + \frac{-3}{7(x-1)}$
2. $\frac{1}{x^2+1} + \frac{x}{(x^2+1)^2}$
3. $\frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$

Answer to Exercise 6 (on page 34)

$$\lim_{x\to -6^-} p(x) = -\infty, \lim_{x\to -6^+} p(x) = \infty$$

$$\lim_{x\to -5^-} p(x) = \lim_{x\to -5^+} p(x) = \lim_{x\to -5} p(x) = 1$$

$$\lim_{x\to -3^-} p(x) = \lim_{x\to -3^+} p(x) = \lim_{x\to -3} p(x) = \frac{1}{3}$$

$$\lim_{x\to\infty} p(x) = 0 \text{ called simply a limit, although it is a left-hand limit}$$

Answer to Exercise 7 (on page 37)

$$\lim_{x\to -\infty} 3^x + 1 = 1; \lim_{x\to 4^+} \log_2(x-4) = -\infty; \lim_{x\to\infty} 2^{1-x} = 0; \lim_{x\to 0^-} \log_{10}(-2x) = -\infty$$

Answer to Exercise 8 (on page 39)

$\lim_{x\to -\infty} \tan^{-1} x = -\frac{\pi}{2}, \lim_{x\to\infty} \tan^{-1} x = \frac{\pi}{2}; \lim_{x\to -\infty} \frac{1}{1+e^{-x}} = 0, \lim_{x\to\infty} \frac{1}{1+e^{-x}} = 1$

Answer to Exercise 9 (on page 41)

1. $\lim_{x\to -1_-} h(x) = 2$ and $\lim_{x\to -1_+} h(x) = 2$, therefore the limit exists and $\lim_{x\to -1} h(x) = 2$

   $\lim_{x\to 0_-} h(x) = 3$ and $\lim_{x\to 0_+} h(x) = 3$, therefore the limit exists and $\lim_{x\to 0} h(x) = 3$

   $\lim_{x\to 1_-} h(x) = 2$ and $\lim_{x\to 1_+} h(x) = 2$, therefore the limit exists and $\lim_{x\to 1} h(x) = 2$

2. $\lim_{x\to -1_-} f(x) = 2$ and $\lim_{x\to -1_+} f(x) = 2$, therefore the limit exists and $\lim_{x\to -1} f(x) = 2$.

   $\lim_{x\to 0_-} f(x) = 3$ and $\lim_{x\to 0_+} f(x) = 0$, and because $\lim_{x\to 0_-} f(x) \neq \lim_{x\to 0_+} f(x)$, the limit does not exist.

$\lim_{x \to 2_-} f(x) = -2$ and $\lim_{x \to 2_+} f(x) = -2$, therefore the limit exists and $\lim_{x \to 2} f(x) = -2$.

3. $\lim_{x \to -2_-} g(x) = -1$ and $\lim_{x \to -2_+} g(x) = -1$, therefore the limit exists and $\lim_{x \to -2} g(x) = -1$.

   $\lim_{x \to 0_-} g(x) = 1$ and $\lim_{x \to 0_+} g(x) = 1$, therefore the limit exists and $\lim_{x \to 0} g(x) = 1$

   $\lim_{x \to 1_-} g(x) = 2$ and $\lim_{x \to 0_+} g(x) = 1$, and because $\lim_{x \to 1_-} g(x) = 2 \neq \lim_{x \to 0_+} g(x)$, the limit does not exist.

   $\lim_{x \to 2_-} g(x) = 0$ and $\lim_{x \to 2_+} g(x) = 0$, therefore the limit exists and $\lim_{x \to 2} g(x) = 0$

Answer to Exercise 10 (on page 44)

1. Never true. If a function is continuous at a, then $f(a) = \lim_{x \to a} f(x)$.
2. Never true. If a function is continuous at a, then $\lim x \to a^+ f(x) = \lim x \to a^- f(x)$
3. Always true. This is the definition of continuity.
4. Sometimes true. The derivative of f at $x = 3$ exists for $f(x) = x^2$ but not for $f(x) = |x - 3|$.
5. Sometimes true. This statement is true for $f(x) = -(x-3)^2$ but not for $f(x) = 4x$.

Answer to Exercise 11 (on page 44)

1. $f(x)$ is not defined at $x = 3$. Therefore, it is also discontinuous at $x = 3$. As we learn about the continuity of polynomials, we will see why $f(x)$ is continuous everywhere else.
2. Here, $f(0)$ is defined, so we need to check if $\lim_{x \to 0} f(x) = f(0)$. The left and right limits as x approaches 0 are the same (∞), so the limit exists. However, $f(0) = 1 \neq \lim_{x \to 0} f(x)$. Therefore, the function is discontinuous at $x = 0$.
3. In this function, $f(3)$ is defined, so we need to check if the limit equals the function value. The limit of $f(x)$ as x approaches 3 is:

$$\lim_{x \to 3} \frac{3x^2 - 8x - 3}{x - 3} = \lim_{x \to 3} \frac{(3x+1)(x-3)}{x-3} = \lim_{x \to 3} 3x + 1 = 10$$

   So the limit exists, but $\lim x \to 3 f(x) \neq f(3)$, and we see that the function is discontinuous at $x = 3$.

Answer to Exercise 13 (on page 48)

1. True, $h(x)$ approaches 2 from the left and right, therefore the limit exists
2. False, $h(x)$ approaches 5 from the left and right, therefore the limit exists
3. False, $h(x)$ approaches 2 from the left and 4 from the right, therefore the limit does not exist
4. True, $\lim_{x \to 5} h(x) = h(5)$, therefore $h(x)$ is continuous at $x = 5$
5. True, $\lim_{x \to 4} h(x)$ does not exist, therefore $h(x)$ is discontinuous at $x = 4$
6. False, $h(2) = 1 \neq 2 = \lim_{x \to 2} h(x)$

Answer to Exercise 13 (on page 48)

1. From the quotient law, we know that:

$$\lim_{x \to 3} \frac{f(x)}{g(x)} = \frac{\lim_{x \to 3} f(x)}{\lim_{x \to 3} g(x)}$$

From the graph, we see that:

$$\lim_{x \to 3} f(x) = -3$$

and that:

$$\lim_{x \to 3} g(x) = 1$$

Substituting these values, we get:

$$\lim_{x \to 3} \frac{f(x)}{g(x)} = \frac{-3}{1} = -3$$

2. From the Sum Law, we know that:

$$\lim_{x \to 2} [f(x) + 5g(x)] = \lim_{x \to 2} f(x) + \lim_{x \to 2} 5g(x)$$

and applying the Constant Multiple Law, we see that:

$$\lim_{x \to 2} [f(x) + 5g(x)] = \lim_{x \to 2} f(x) + 5 \lim_{x \to 2} g(x)$$

Examining the graph of $f(x)$ and $g(x)$, we can determine that

$$\lim_{x \to 2} f(x) = -2$$

and

$$\lim_{x/to2} g(x) = 0$$

Substituting these values, we get:

$$\lim_{x \to 2} [f(x) + 5g(x)] = -2 + 5 \cdot 0 = -2$$

3. From the quotient law, we see that:

$$\lim_{x \to -1} \frac{3g(x)}{f(x)} = \frac{\lim_{x \to -1} 3g(x)}{\lim_{x \to -1} f(x)}$$

Applying the Constant Multiple Law, we get:

$$\lim_{x \to -1} \left[\frac{3g(x)}{f(x)}\right] = \frac{3 \lim_{x \to -1} g(x)}{\lim_{x \to -1} f(x)}$$

From the graph, we see that:

$$\lim_{x \to -1} f(x) = 2$$

and

$$\lim_{x \to -1} g(x) = 2$$

Substituting, we get:

$$\lim_{x \to -1} \left[\frac{3g(x)}{f(x)}\right] = \frac{3 \cdot 2}{2} = 3$$

4. Applying the Product and Constant Multiple Laws, we get:

$$\lim_{x \to 0} [f(x) \cdot 5g(x)] = \lim_{x \to 0} f(x) \cdot 5 \cdot \lim_{x \to 0} g(x)$$

Examining the graphs, we see that $\lim_{x \to 0} f(x)$ does not exist and $\lim_{x \to 0} g(x) = 1$. Because $\lim_{x \to 0} f(x)$ does not exist, $\lim_{x \to 0} f(x) \cdot 5 \cdot \lim_{x \to 0} g(x)$ also does not exist.

5. Applying the Difference and Constant Multiple Laws, we see that:

$$\lim_{x\to -1}[f(x) - 3g(x)] = \lim_{x\to -1} f(x) - 3\cdot \lim_{x\to -1} g(x)$$

Examining the graphs, we see that:

$$\lim_{x\to -1} f(x) = 2$$

and

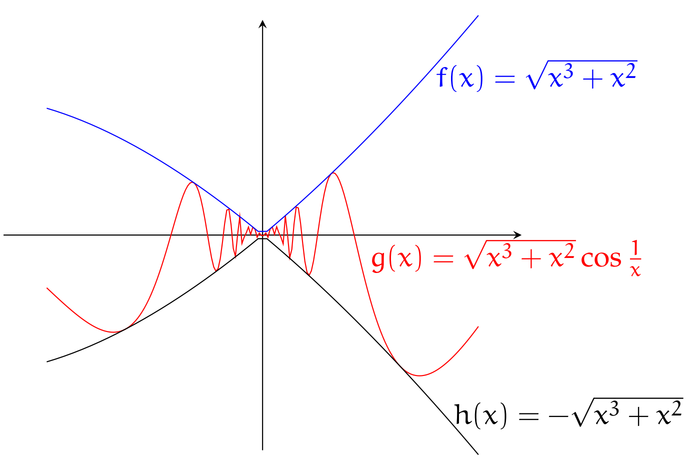
$$\lim_{x\to -1} g(x) = 2$$

Substituting, we get that:

$$\lim_{x\to -1}[f(x) - 3g(x)] = 2 - 3\cdot 2 = 2 - 6 = -4$$

Answer to Exercise 14 (on page 51)

Let $f(x) = -\sqrt{x^3+x^2}$ and $h(x) = \sqrt{x^3+x^2}$. Near 0, $f(x) \le \sqrt{x^3+x^2}\cos\frac{1}{x} \le h(x)$. Additionally, $\lim_{x\to 0} f(x) = \lim_{x\to 0} h(x) = 0$. Therefore, by the Squeeze Theorem, we can state that $\lim_{x\to 0}\sqrt{x^3+x^2}\cos\frac{1}{x} = 0$. Plotting all three functions, we can confirm our answer:



Answer to Exercise 15 (on page 52)

The question tells us that the function in question, $f(x)$, is between two other functions.

$\lim_{x\to 2} 2x+3 = 2(2)+3 = 7$ and $\lim_{x\to 2} x^2 - 2x + 7 = 2^2 - 2(2) + 7 = 7$. Since the limits are equal, by Squeeze Theorem we can also know that $\lim_{x\to 2} f(x) = 7$.

Answer to Exercise 16 (on page 52)

Note that we can only evaluate the limit from the right, as the domain for this function is $x \geq 0$. Since the range of the sine function is $[-1, 1]$, we can state that

$$-1 \leq \sin\frac{\pi}{x} \leq 1$$

and therefore

$$\frac{1}{e} \leq e^{\sin\frac{\pi}{x}} \leq e$$

. Because we assume the positive root, it is also true that

$$\frac{\sqrt{x}}{e} \leq \sqrt{x}e^{\sin\frac{\pi}{x}} \leq \sqrt{x}e$$

. Taking the limits of the border functions, we see that

$$\lim_{x\to 0^+} \frac{\sqrt{x}}{e} = \lim_{x\to 0^+} \sqrt{x}e = 0$$

Therefore,

$$\lim_{x\to 0^+} \sqrt{x}e^{\sin\frac{\pi}{x}} = 0$$

Answer to Exercise 17 (on page 54)

1. define $f(x) = 2x^4 + x - 12$, $a = 1$, $b = 2$, and $N = 0$. Calculate $f(a)$ and $f(b)$:

   $$f(1) = -9 \text{ and } f(2) = 22$$

   Since $f(x)$ is a polynomial, it is continuous on the interval $x \in [1, 2]$ and we see that $f(a) < 0 < f(b)$. Therefore, there exists some $c \in [1, 2]$ such that $f(c) = 0$.

2. First, we can rearrange the equation we are considering and define $f(x) = \ln x - 3x + 4\sqrt{x}$, and realize we are looking for values where $f(x) = 0$. Both $\ln x$ and $\sqrt{x}$ are only continuous for $x > 0$. The interval we are interested in, $x \in [2, 3]$, is in the domain of continuity for both $\ln x$ and $\sqrt{x}$. Defining $a = 2$, $b = 3$, and $N = 0$, we find that $f(a) = 0.35 > N > -0.973 = f(b)$. Since $N \in [f(b), f(a)]$, there must exist some c such that $f(c) = N = 0$ and there is a solution to the equation $\ln x = 3x - 4\sqrt{x}$ on the interval $x \in (2, 3)$.

3. Similar to above, define $f(x) = 2\sin x - 3x^2 + 2x$, $a = 1$, $b = 2$, and $N = 0$. Calculate that $f(a) = 0.683$ and $f(b) = -6.181$. Since $f(x)$ is continuous on the interval $x \in [1, 2]$ and $f(b) < N < f(a)$, there exists some $c \in (1, 2)$ such that $f(c) = 0$. Therefore, there is a solution to $\sin x = 3x^2 - 2x$ on the interval $x \in (1, 2)$.

Answer to Exercise 18 (on page 58)

To sketch the curve, we begin by choosing several values of t and computing the corresponding x- and y-values.

t	x	y
-2	-1	4
-1	0	1
0	1	0
1	2	1
2	3	4

Plotting these points produces a parabola opening upward. As t increases, the x-values increase and the curve is traced from left to right.

Although the rectangular equation of the curve is

$$y = (x - 1)^2,$$

the parametric equations determine the direction in which the curve is drawn, which cannot be seen from the rectangular form alone.

Answer to Exercise 19 (on page 61)

To sketch the curve, we begin by evaluating the parametric equations at several values of t.

t	x	y
0	4	3
1	2	5
2	0	-1
3	-2	-15

Plotting these points produces a curve that opens downward. As t increases, the x-values decrease, indicating that the curve is traced from right to left.

The Python animation confirms this behavior by showing the curve being drawn progressively as t increases. The moving point illustrates both the direction of motion and how the curve is traced over time, reinforcing the hand-drawn sketch.

Answer to Exercise 20 (on page 64)

First compute the derivatives:

$$\frac{dx}{dt} = 3t^2 - 3, \quad \frac{dy}{dt} = 2t.$$

Horizontal tangents occur when $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$.

$$2t = 0 \Rightarrow t = 0.$$

At $t = 0$, the point on the curve is

$$x = 0, \quad y = 0.$$

Vertical tangents occur when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$.

$$3t^2 - 3 = 0 \Rightarrow t = \pm 1.$$

At $t = 1$, the point is

$$(-2, 1),$$

and at $t = -1$, the point is

$$(2, 1).$$

Answer to Exercise 21 (on page 65)

First compute the derivatives with respect to t:

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 2t.$$

The derivative of the curve is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{2} = t.$$

At $t = 1$,

$$\frac{dy}{dx} = 1.$$

Thus, the slope of the tangent line at the corresponding point on the curve is 1.

Answer to Exercise 22 (on page 68)

Two sets of parametric equations can represent the same graph but describe different curves because the parameter controls how the graph is traced, not just the shape of the graph itself.

Different parametric equations may generate the same set of points in the plane while differing in one or more of the following ways:

- the direction of motion along the curve,
- the speed at which the curve is traced,
- the starting point of the curve,
- or the number of times the curve is traced.

Answer to Exercise 23 (on page 78)

The radius is 5 m, because the coefficients of both the x and y functions is 5.

Recall that the velocity in each direction is given by the derivative of the position functions:

$$v_x(t) = \frac{d}{dt}[5\cos(3t)] = -15\sin(3t)$$

$$v_y(t) = \frac{d}{dt}[5\sin(3t)] = 15\cos(3t)$$

The overall *speed* can be found from the x and y components of the velocity:

$$
\begin{aligned}
|v| &= \sqrt{v_x^2 + v_y^2} \\
&= \sqrt{[-15\sin(3t)]^2 + [15\cos(3t)]^2} \\
&= \sqrt{15^2\left[\sin(3t)^2 + \cos(3t)^2\right]} \\
&= \sqrt{15^2} = 15\frac{m}{s}
\end{aligned}
$$

Notice that this is the coefficient for both components of the velocity.

To complete the path, the particle must travel 10π m. If the speed is $15\frac{m}{s}$, then the time it takes is:

$$t = \frac{d}{v} = \frac{10\pi \text{ m}}{15\frac{m}{s}} = \frac{2\pi}{3}\text{s} \approx 2.09 \text{ s}$$

Answer to Exercise 24 (on page 84)

$$\frac{120 \text{ km}}{1\text{hour}} = \frac{1000 \text{ m}}{1 \text{ km}}\frac{120 \text{ km}}{1\text{hour}}\frac{1 \text{ hour}}{3600 \text{ seconds}} = 33.3 \text{ m/s}$$

$$F = \frac{mv^2}{r} = \frac{0.4(33.3)^2}{200} = 2.2 \text{ newtons}$$

Answer to Exercise 25 (on page 85)

Givens:

$$
\begin{gathered}
T_{max} = (20\text{kg}) \cdot \left(9.8\frac{m}{s^2}\right) = 196\text{N} \\
r = 0.35m \\
m_{whistle} = 165g = 0.165kg
\end{gathered}
$$

Unknown:

$$
\begin{gathered}
v = ? \\
f = ?
\end{gathered}
$$

Equation(s):

$$F = ma$$
$$a = \frac{v^2}{r}$$
$$v = rf$$

Solution: First, we find the tangential speed if the tension in the string is the maximum tension:

$$T_{max} = m_{whistle} a = m_{whistle} \frac{v^2}{r}$$
$$v = \sqrt{\frac{T_{max} r}{m_{whistle}}}$$
$$v = \sqrt{\frac{196\text{N}\,(0.35\text{m})}{0.165\text{kg}}} \approx 0.645 \frac{\text{m}}{\text{s}}$$

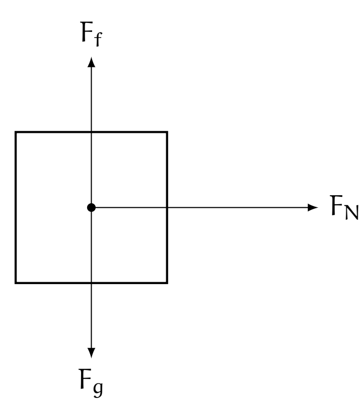
Therefore, the maximum tangential speed of the whistle before the string breaks is $0.645 \frac{m}{s}$. Now finding the equivalent frequency (rotations per second):

$$f = \frac{v}{r} = \frac{0.645 \frac{m}{s}}{0.35m} \approx 1.842\text{Hz}$$

So, to break the string, the lifeguard would have to spin it at nearly 2 rotations per second, which is achievable. The lifeguard could possibly break the string.

Answer to Exercise 26 (on page 85)

There are three forces acting on the rider: gravity, friction with the wall, and the normal force with the wall:



The FBD lets us write equations for Newton's Second Law in each dimension:

$$(1)\;\; ma_x = F_N$$

$$(2)\;\; ma_y = F_f - F_g = \mu F_N - mg$$

Because the rider doesn't fall down, we know that $a_y = 0\frac{m}{s^2}$ and therefore equation (2) becomes:

$$\mu F_N - mg = 0 \rightarrow F_N = \frac{mg}{\mu}$$

Having solved for F_N, we substitute for it into equation (1):

$$ma_x = \frac{mg}{\mu} \rightarrow a_x = \frac{g}{\mu}$$

Since we know g and μ, we can calculate a_x:

$$a_x = \frac{9.8\frac{m}{s^2}}{0.32} = 30.625\frac{m}{s^2}$$

This is the minimum acceleration needed to keep the rider from slipping down. We can now use the relationship between centripetal acceleration, tangential velocity, and the radius to find the angular velocity:

$$a = \frac{v^2}{r} = \frac{(\omega r)^2}{r} = \omega^2 r$$

$$\omega = \sqrt{\frac{a}{r}} = \sqrt{\frac{30.625\frac{m}{s^2}}{10\text{ m}}} \approx 1.75\frac{rad}{s}$$

Answer to Exercise 27 (on page 88)

Givens:

- Mass of tetherball: $m = 0.5$ kg
- Length of string: $L = 2.5$ m
- Angle with vertical: $\theta = 38°$

The components of the tension force are separated as follows:

$$T\cos\theta = mg, \qquad T\sin\theta = \frac{mv^2}{r}$$

Dividing the second equation by the first equation gives:

$$\tan\theta = \frac{v^2}{rg} \rightarrow v = \sqrt{rg\tan\theta}$$

But we are not given r. We can find r using trigonometry:

$$r = L\sin\theta$$

Now we can find v:

$$v = \sqrt{(L\sin\theta)g\tan\theta}$$

Solving this gives us:

$$v = \sqrt{(2.5\text{ m}\sin 38^\circ)(9.8\frac{\text{m}}{\text{s}^2})\tan 38^\circ} \approx 3.4\frac{\text{m}}{\text{s}}$$

The speed of the tetherball is approximately $3.4\frac{\text{m}}{\text{s}}$.

Answer to Exercise 28 (on page 92)

- At the top of the loop: At the top of the loop, both the gravitational force and the normal force point downward. Therefore, we can write:

  $$F_{net} = F_N + F_g = \frac{mv^2}{r}$$

  Solving for the normal force gives:

  $$F_N = \frac{mv^2}{r} - F_g$$

  Substituting for $F_g = mg$ and $r = \frac{d}{2} = 25$ m:

  $$F_N = \frac{(1200\text{ kg})(25\frac{\text{m}}{\text{s}})^2}{25\text{ m}} - (1200\text{ kg})(10\frac{\text{m}}{\text{s}^2})$$

  $$F_N = 30000\text{ N} - 12000\text{ N} = 18000\text{ N}$$

- At the bottom of the loop: At the bottom of the loop, the gravitational force points downward while the normal force points upward. Therefore, we can write:

  $$F_{net} = F_N - F_g = \frac{mv^2}{r}$$

Solving for the normal force gives:

$$F_N = \frac{mv^2}{r} + F_g$$

Substituting for $F_g = mg$ and $r = \frac{d}{2} = 25$ m:

$$F_N = \frac{(1200 \text{ kg})(25\frac{m}{s})^2}{25 \text{ m}} + (1200 \text{ kg})(10\frac{m}{s^2})$$

$$F_N = 30000 \text{ N} + 12000 \text{ N} = 42000 \text{ N}$$

Answer to Exercise 29 (on page 98)

The correct answer is: The frequency would increase by a factor of $\sqrt{2}$.

The frequency of oscillation for a mass-spring system is given by the formula:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

where k is the spring constant and m is the mass attached to the spring. Since frequency is inversely proportional to the square root of the mass, if the mass is halved, the new frequency f' can be calculated as follows:

$$\begin{aligned} f' &= \frac{1}{2\pi}\sqrt{\frac{k}{m/2}} \\ &= \frac{1}{2\pi}\sqrt{\frac{2k}{m}} \\ &= \sqrt{2} \cdot \frac{1}{2\pi}\sqrt{\frac{k}{m}} \\ &= \sqrt{2} \cdot f \end{aligned}$$

Therefore, the frequency increases by a factor of $\sqrt{2}$ when the

Answer to Exercise 30 (on page 100)

First, we find the angular frequency ω using the formula:

$$w = 2\pi f = \sqrt{\frac{k}{m}} = \sqrt{\frac{400\,\text{N/m}}{4\,\text{kg}}} = \sqrt{100} = 10\,\text{rad/s}$$

Therefore, the general equation for the position of a mass-spring system undergoing simple harmonic motion is:

$$x(t) = A\cos(\omega t + \phi)$$

becomes $x = 6\sin(10t + \phi)$ since the initial velocity is 0, meaning it starts at maximum displacement. Solving for ϕ using the initial condition $x(0) = 6$ cm:

$$6 = 6\sin(\phi) \implies \sin(\phi) = 1 \implies \phi = \frac{\pi}{2}$$

So the base equation becomes:

$$x(t) = 6\sin(10t + \frac{\pi}{2})$$

Which is option C.

Answer to Exercise 31 (on page 101)

To find the speed of the ball as it leaves the spring, we can use the principle of conservation of mechanical energy. The net energy in the system remains constant, meaning that the potential energy stored in the compressed spring is converted into the kinetic energy of the ball when it is released. The potential energy stored in the spring when compressed is given by:

$$U_s = \frac{1}{2}kx^2 = \frac{1}{2}(500)(-0.24)^2 = 14.4\,\text{J}$$

The kinetic energy of the ball as it leaves the spring is the same amount, so we set the potential energy equal to the kinetic energy:

$$\frac{1}{2}mv^2 = U_s$$

Solving for v:

$$v = \sqrt{\frac{2U_s}{m}} = \sqrt{\frac{2(14.4)}{0.2}} = \sqrt{144} = 12\,\text{m/s}$$

The speed of the ball as it leaves the spring is $12\,\text{m/s}$.

Answer to Exercise 32 (on page 103)

The correct differential equation is $x'' + 100x = 0$. This is derived from the equation of motion for any simple harmonic oscillator: $mx'' + kx = 0$, where $m = 2\,\text{kg}$ and $k = 200\,\text{N/m}$. Dividing by m, we get $x'' + \frac{k}{m}x = 0$, which simplifies to $x'' + 100x = 0$.

Answer to Exercise 33 (on page 104)

- This system is damped. Specifically, it is underdamped because the amplitude of oscillation decreases over time.
- To make this system undergo simple harmonic motion, we would need to remove the damping term from the equation of motion. This means setting the damping coefficient c to zero in the differential equation, which would eliminate the exponential decay factor in the position function. The equation becomes $x(t) = 0.15\cancel{e^{-0.40t}}\cos(4t + 0.30) \rightarrow x(t) = 0.15\cos(4t + 0.30)$

Answer to Exercise 34 (on page 105)

1. Notice that the system equation is linear as it is defined with an x'' and x.The general solution to the differential equation $x'' + 16x = 0$ is of the form $x(t) = A\cos(4t) + B\sin(4t)$. This comes from the idea that the characteristic equation $r^2 + 16 = 0$ has roots $r = \pm 4i$, where the β term is 4. Using the initial conditions $x(0) = 0.20$ and $v(0) = x'(0) = 0$, we find our velocity equation to be $v(t) = x'(t) = -4A\sin(4t) + 4B\cos(4t)$.

   From $x(0) = A\cos 0 + B\sin 0 = A(1) + B(0) = A = 0.20$, we find that $A = 0.20$.

   From $v(0) = -4A\sin 0 + 4B\cos 0 = -4(0.20)(0) + 4B(1) = 4B = 0$, we find that $B = 0$.

   Therefore, the solution is:
   $$x(t) = 0.20\cos(4t)$$

2. To find the acceleration, we double differentiate $x(t)$ with respect to time: To find the velocity, we differentiate $x(t)$ with respect to time:

   $$v(t) = x'(t) = -0.80\sin(4t)$$

   $$a(t) = v'(t) = -3.20\cos(4t)$$

   Since cosine and sine functions oscillate between -1 and 1 (regardless of angular frequency $\omega = 4$), the maximum velocity is 0.80 m/s and the maximum acceleration is $3.20\,\text{m/s}^2$.

Answer to Exercise 35 (on page 107)

We know that the restoring force for a pendulum is given by $F = -mg\sin\theta$. For small angles, we can approximate $\sin\theta \approx \theta$ (in radians), leading to $F \approx -mg\theta$. For any point on the circle, we can relate the angular displacement θ to the arc length x using $x = L\theta$, where L is the length of the pendulum. Thus, we can rewrite the restoring force in terms of the second derivative, which represents angular acceleration:

$$a_{tan} = \frac{d^2x}{dt^2} = L\frac{d^2\theta}{dt^2} \tag{9.9}$$

Using Newton's Second Law, we have:

$$mL\frac{d^2\theta}{dt^2} = -mg\sin\theta \tag{9.10}$$

$$L\frac{d^2\theta}{dt^2} + g\sin\theta = 0 \tag{9.11}$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0 \tag{9.12}$$

But for small angles, we can approximate $\sin\theta \approx \theta$, leading to:

$$[label = eq : linearpendulum]\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \tag{9.13}$$

This is a second-order linear differential equation with constant coefficients, similar to the mass-spring system problems discussed earlier. The angular frequency ω of the pendulum can be identified as:

$$\omega = \sqrt{\frac{g}{L}} \tag{9.14}$$

Therefore, the period T of the pendulum is given by:

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{9.15}$$

Answer to Exercise 36 (on page 108)

The correct answer is: Approximately 4.4 seconds.

The period T of a simple pendulum is given by the formula:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where L is the length of the pendulum and g is the acceleration due to gravity.

On Earth, the period is 2 seconds, so we can set up the equation:

$$2 = 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}}$$

This shows that T is inversely proportional to the square root of g. We can rearrange this equation to solve for L. So if g decreases by a factor of 6, then T will increase by a factor of $\sqrt{6}$.

Now, on the Moon, the acceleration due to gravity is approximately $\frac{1}{6}g_{\text{Earth}}$. We can substitute this into the period formula:

$$T_{\text{Moon}} = 2\pi\sqrt{\frac{L}{g_{\text{Moon}}}} = 2\pi\sqrt{\frac{L}{\frac{1}{6}g_{\text{Earth}}}} = 2\pi\sqrt{\frac{6L}{g_{\text{Earth}}}} = \sqrt{6}\cdot 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}} = \sqrt{6}\cdot T_{\text{Earth}}$$

Since our original period on Earth is 2 seconds, we have:

$$T_{\text{Moon}} = \sqrt{6}\cdot 2 \approx 4.9 \text{ seconds}$$

Which is equal to option (c).

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